# Metrics review, part 2

*EC421, Set 03*

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*Prologue*

# R showcase

## `ggplot2`

- Powerful graphing and mapping package for R.
- *Idea:* Build your figures layer by layer.
- Exportable to many applications; part of the `tidyverse`.

## `shiny`

- Export your figures and code to interactive web apps.
- Enormous range of applications
  - ↳ SuperZips
  - ↳ Retirement simulation
  - ↳ Distribution calculator

# Schedule

*Last Time*

We reviewed the fundamentals of statistics and econometrics.

*Today*

We review more of the main/basic results in metrics.

*This week*

We will post the **first assignment** (focused on *review*) soon.
First we need to finish more (of this) review.

*Multiple regression*

# Multiple regression

*More explanatory variables*

We're moving from **simple linear regression**
(one outcome variable and one explanatory variable)

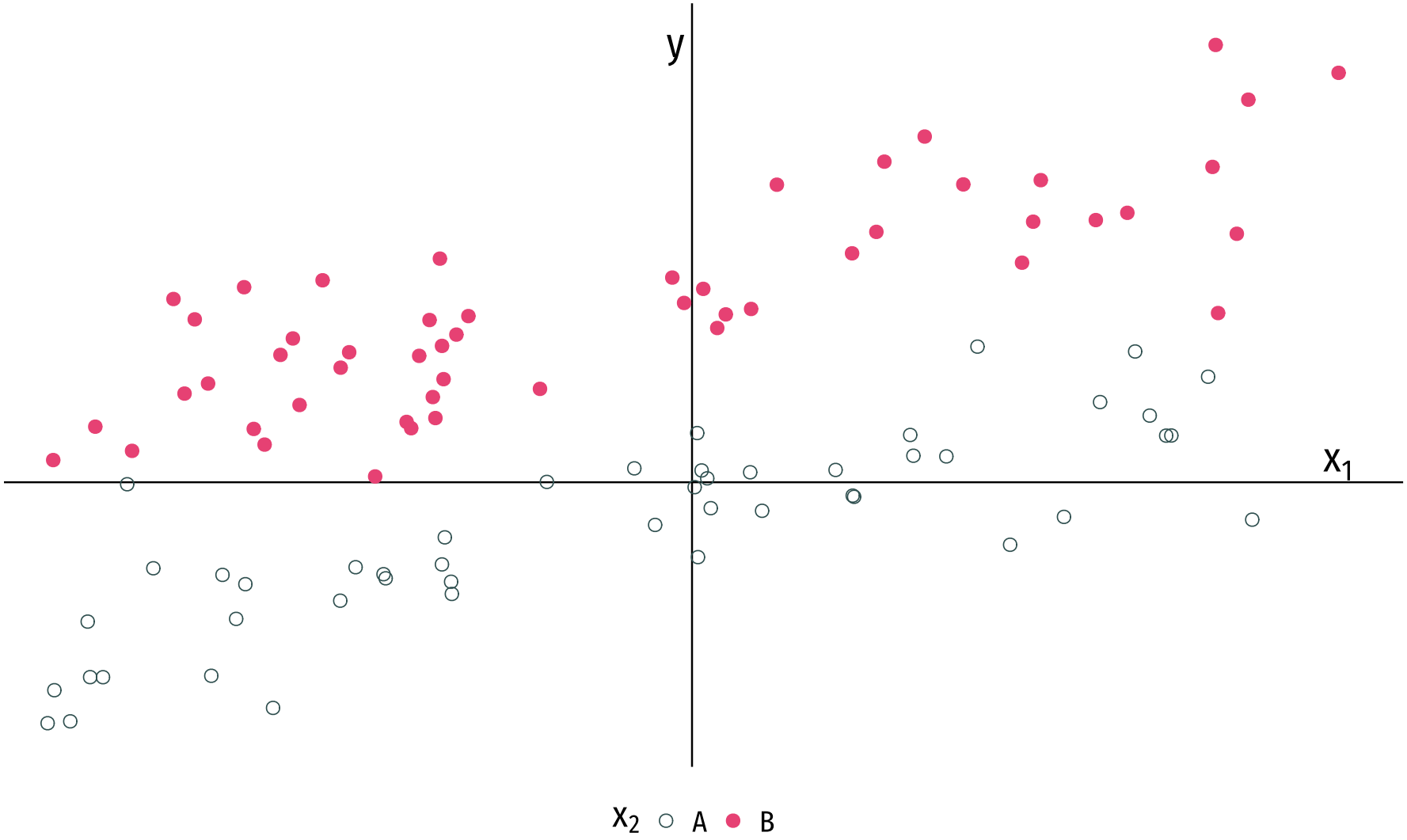
$$y_i = \beta_0 + \beta_1 x_i + u_i$$

to the land of **multiple linear regression**
(one outcome variable and multiple explanatory variables)

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_k x_{ki} + u_i$$

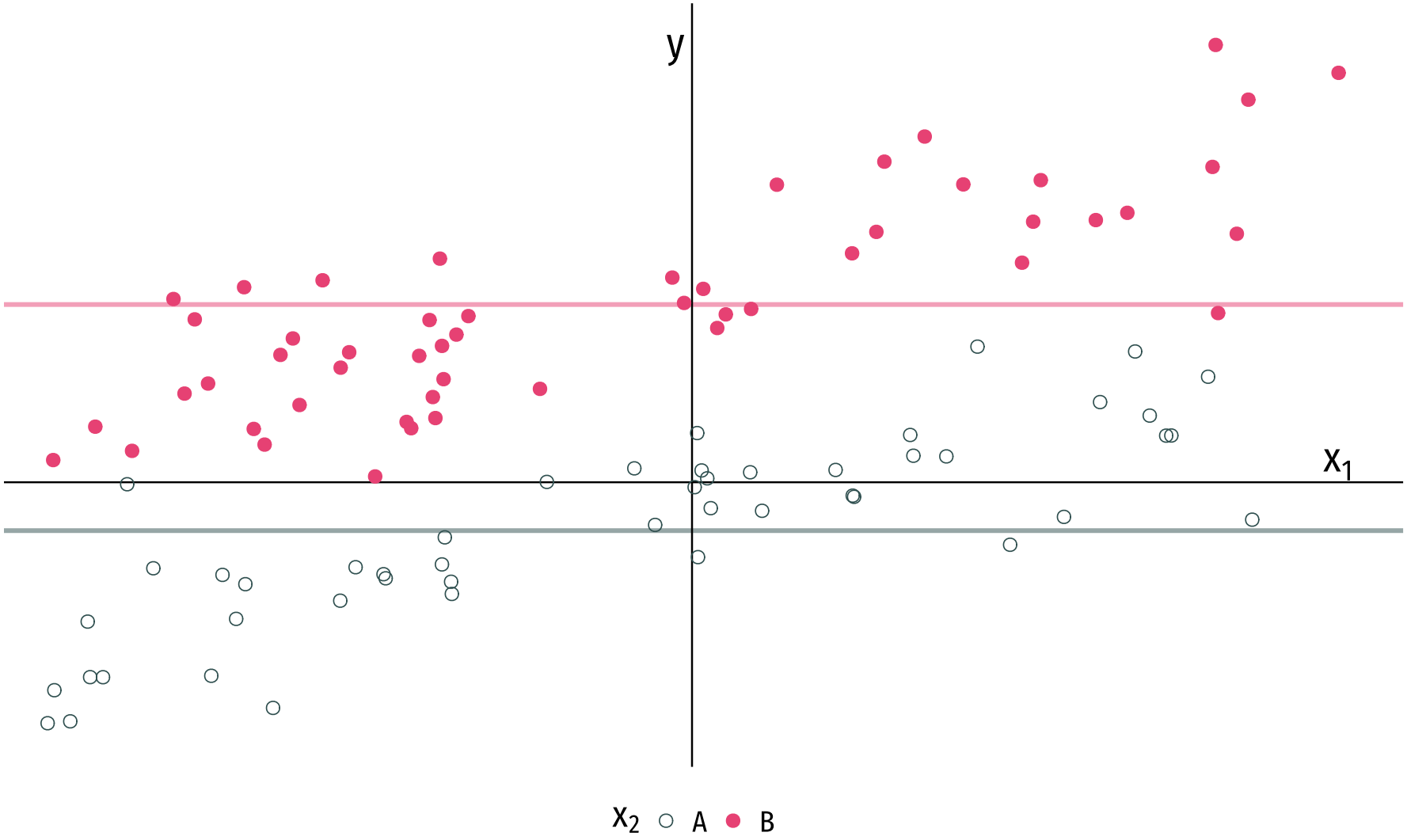
# Multiple regression

$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i$ $x_1$ is continuous $x_2$ is categorical



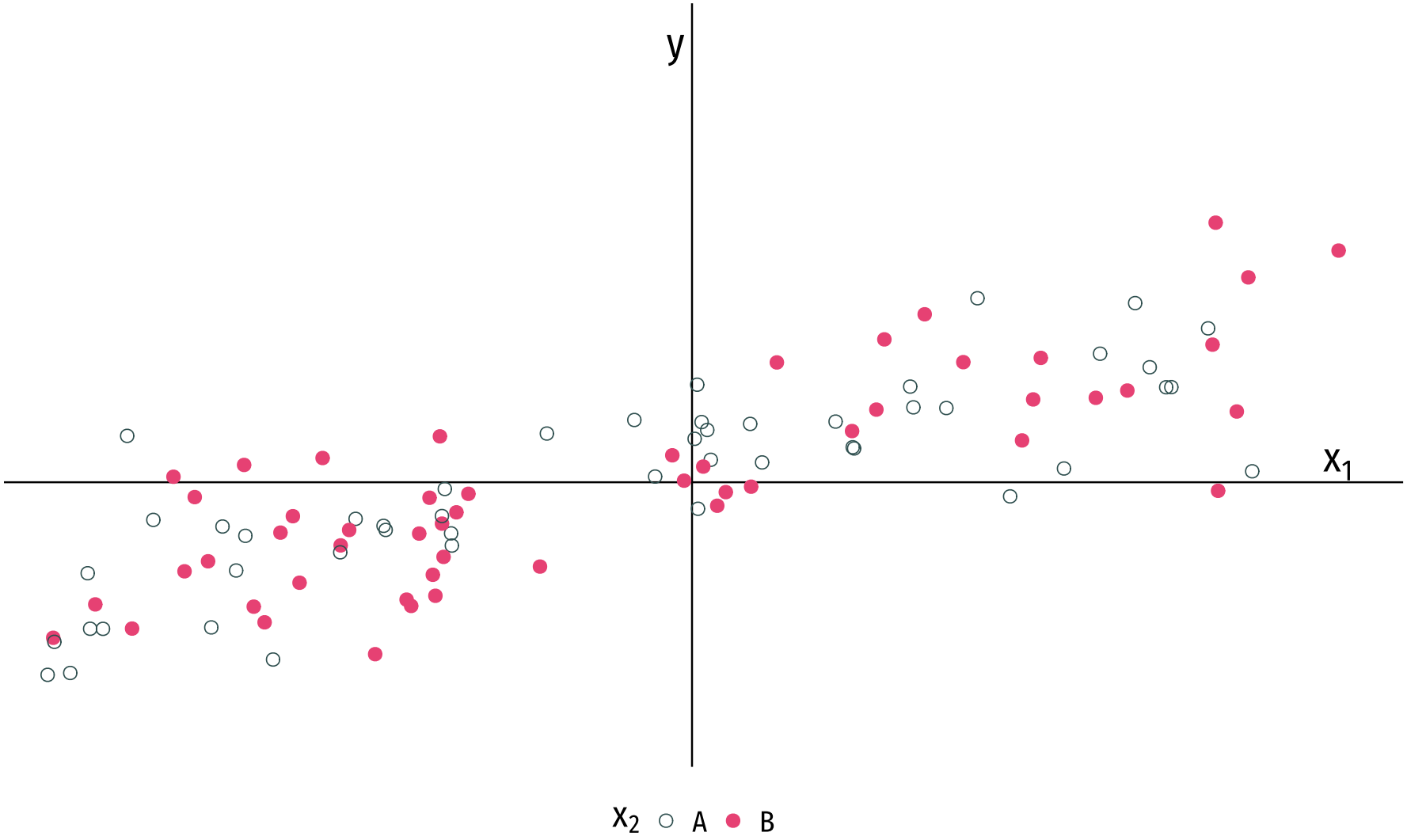
# Multiple regression

The intercept and categorical variable $x_2$ control for the groups' means.



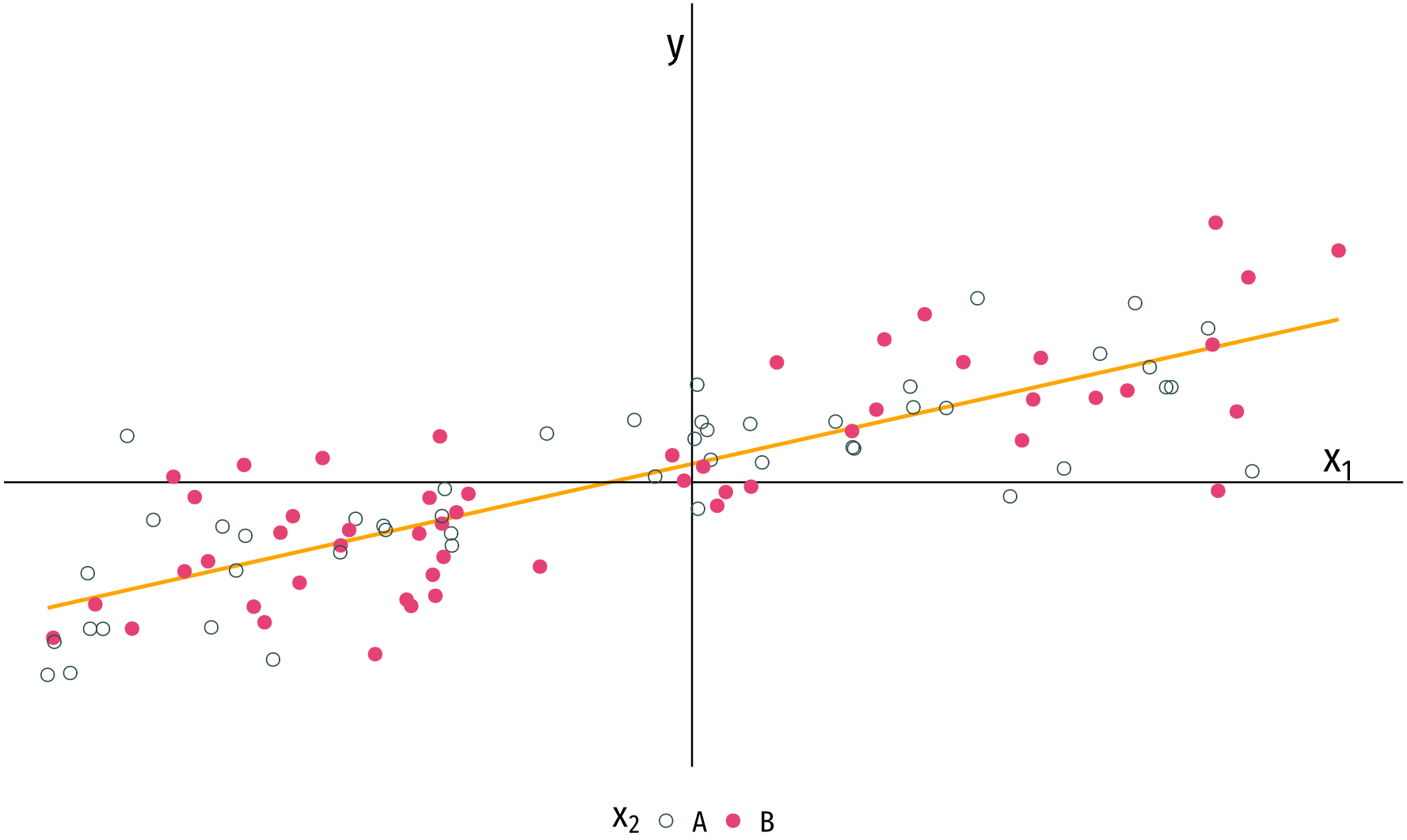
# Multiple regression

With groups' means removed:



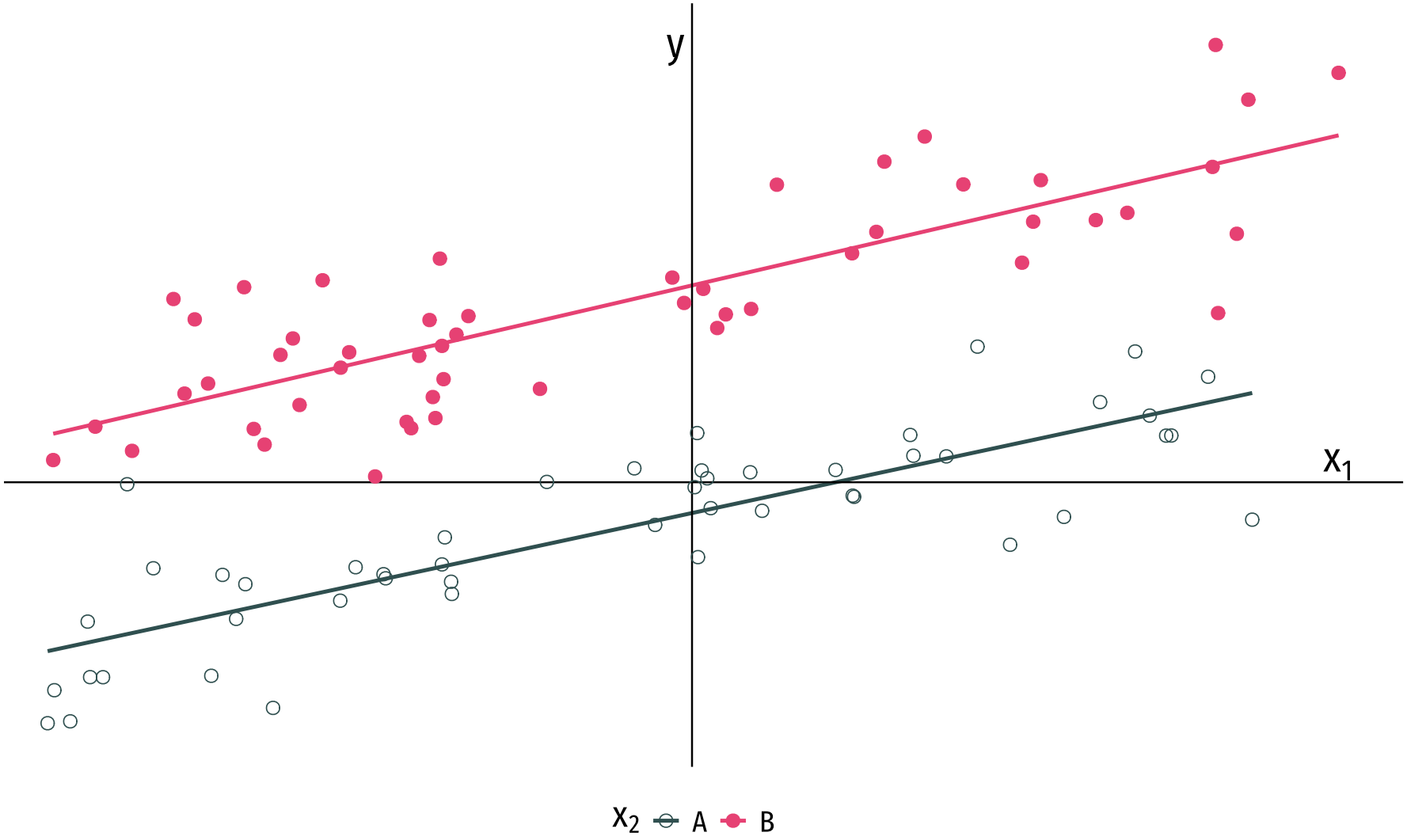
# Multiple regression

$\hat{\beta}_1$ estimates the relationship between $y$ and $x_1$ after controlling for $x_2$.



# Multiple regression

Another way to think about it: We're estimating two (parallel) lines.



Looking at our estimator can also help.

# Multiple regression

For the simple linear regression $y_i = \beta_0 + \beta_1 x_i + u_i$

$$
\begin{aligned}
\hat{\beta}_1 &= \\
&= \frac{\sum_i (x_i - \overline{x})(y_i - \overline{y})}{\sum_i (x_i - \overline{x})} \\
&= \frac{\sum_i (x_i - \overline{x})(y_i - \overline{y})/(n-1)}{\sum_i (x_i - \overline{x})/(n-1)} \\
&= \frac{\hat{\mathrm{Cov}}(x, y)}{\hat{\mathrm{Var}}(x)}
\end{aligned}
$$

# Multiple regression

Simple linear regression estimator:

$$\hat{\beta}_1 = \frac{\hat{\mathrm{Cov}}(x,\, y)}{\hat{\mathrm{Var}}\,(x)}$$

# Multiple regression

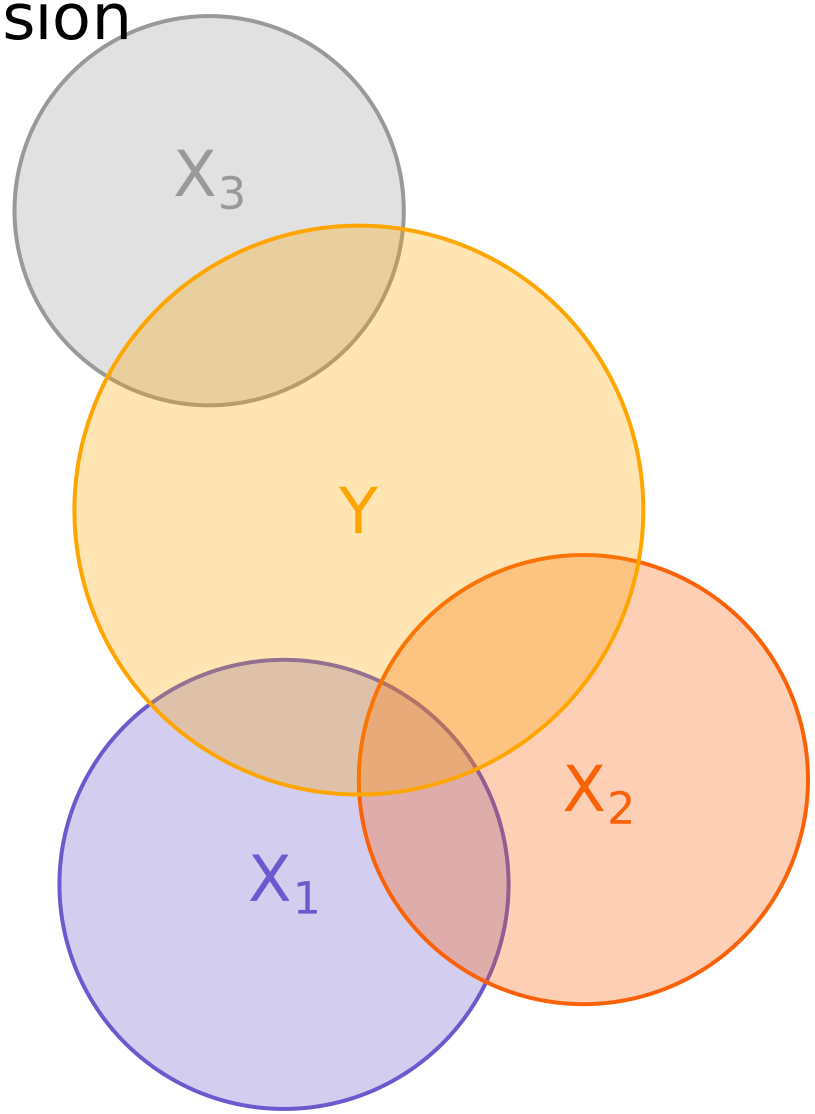
Consider the multiple-regression model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + u$$

*Residualized* $x_1$ ($\tilde{x}_1$) comes from regressing $x_1$ on an intercept and *all other explanatory variables* (then collecting the residuals), *i.e.*,

$$\begin{aligned} \hat{x}_1 &= \hat{\gamma}_0 + \hat{\gamma}_2\, x_2 + \hat{\gamma}_3\, x_3 \\ \tilde{x}_1 &= x_1 - \hat{x}_1 \end{aligned}$$

# Multiple regression



# Multiple regression

*Model fit*

Measures of *goodness of fit* quantify how well a model *describes/fits* the data.

**Common measure:** $R^2$ [R-squared] (*a.k.a.* coefficient of determination)

$$R^2 = \frac{\sum_i (\hat{y}_i - \overline{y})^2}{\sum_i (y_i - \overline{y})^2} = 1 - \frac{\sum_i \left(y_i - \hat{y}_i\right)^2}{\sum_i (y_i - \overline{y})^2}$$

Notice our old friend SSE: $\sum_i \left(y_i - \hat{y}_i\right)^2 = \sum_i e_i^2$.

# Multiple regression

**The problem:** As we add variables to our model, $R^2$ *mechanically* increases.

# Multiple regression

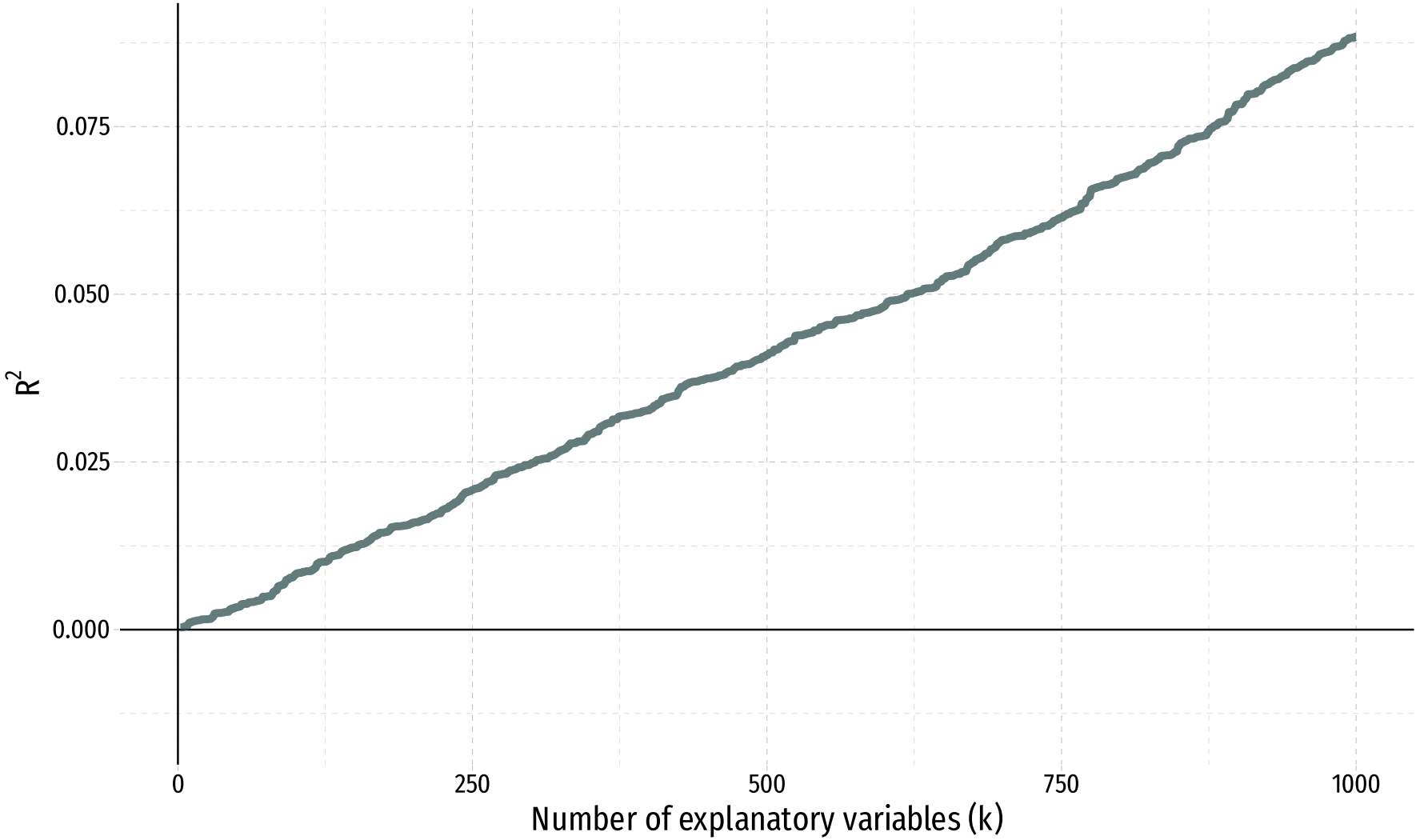
**The problem:** As we add variables to our model, $R^2$ *mechanically* increases.

R code for the simulation:

```
set.seed(1234)
y = rnorm(1e4)
x = matrix(data = rnorm(1e7), nrow = 1e4)
x %<>% cbind(matrix(data = 1, nrow = 1e4, ncol = 1), x)
r_df = mclapply(X = 1:(1e3-1), mc.cores = detectCores() - 1, FUN = function(i) {
  tmp_reg = lm(y ~ x[,1:(i+1)]) %>% summary()
  data.frame(
    k = i + 1,
    r2 = tmp_reg %$% r.squared,
    r2_adj = tmp_reg %$% adj.r.squared
  )
}) %>% bind_rows()
```

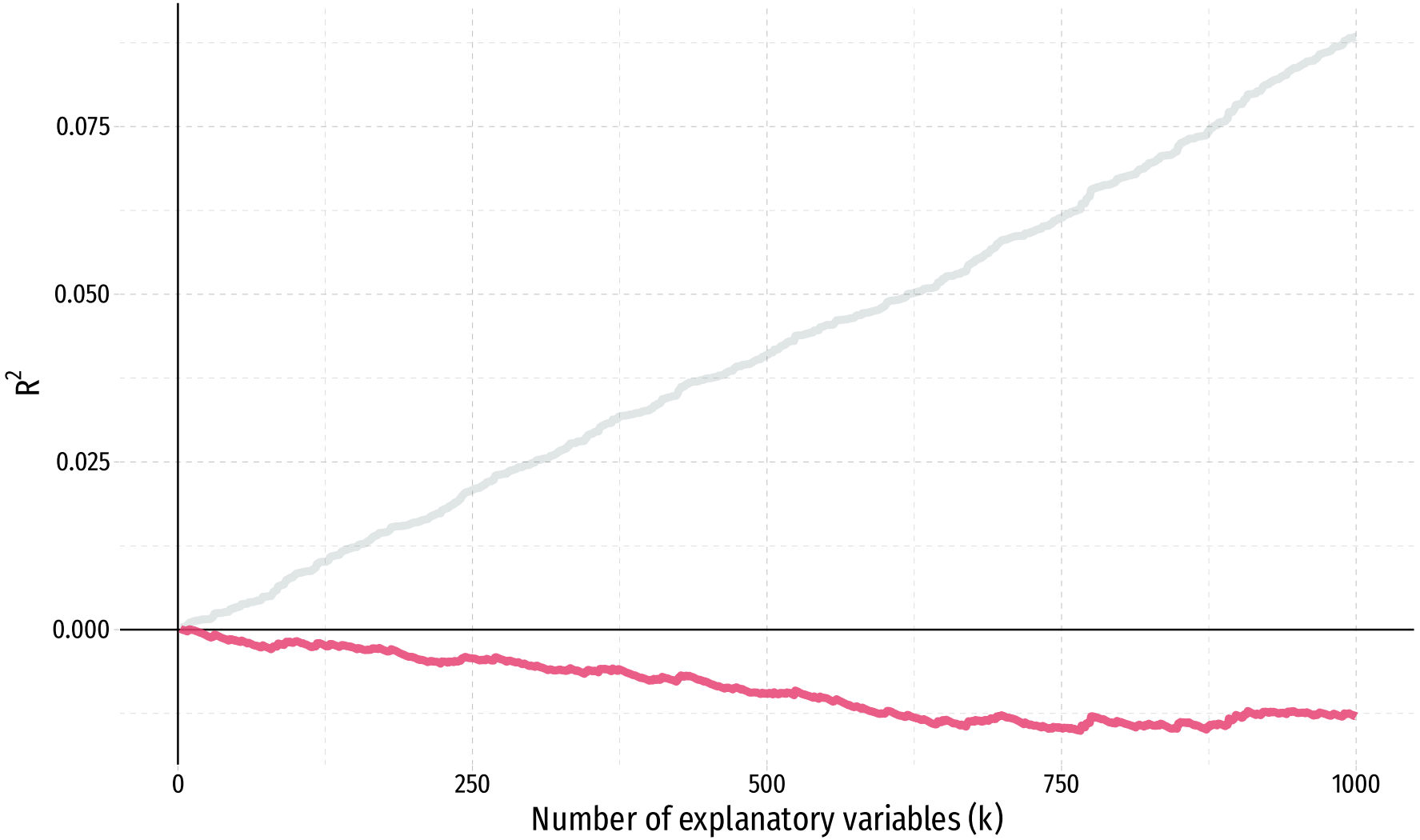
# Multiple regression

**The problem:** As we add variables to our model, $R^2$ *mechanically* increases.



# Multiple regression

## **One solution:** Adjusted $R^2$



# Multiple regression

**The problem:** As we add variables to our model, $R^2$ *mechanically* increases.

**One solution:** Penalize for the number of variables, *e.g.*, adjusted $R^2$:

$$\overline{R}^2 = 1 - \frac{\sum_i \left(y_i - \hat{y}_i\right)^2 / (n-k-1)}{\sum_i \left(y_i - \overline{y}\right)^2 / (n-1)}$$

*Note:* Adjusted $R^2$ need not be between 0 and 1.

# Multiple regression

*Tradeoffs*

There are tradeoffs to remember as we add/remove variables:

**Fewer variables**

- generally explain less variation in $y$,
- provide simple interpretations and visualizations (*parsimonious*),
- may need to worry about omitted-variable bias (OVB).

**More variables**

- more likely to find *spurious* relationships (statistically significant due to chance—does not reflect a true, population-level relationship),
- more difficult to interpret the model,
- may still miss important variabless—still OVB.

# Omitted-variable bias

# Omitted-variable bias

We'll go deeper into this issue in a few weeks, but as a refresher:

**Omitted-variable bias** (OVB) arises when we omit a variable that

1. affects our outcome variable $y$
2. correlates with an explanatory variable $x_j$

As it's name suggests, this situation leads to bias in our estimate of $\beta_j$.

# Omitted-variable bias

## Example

Let's imagine a simple model of the *returns to schooling*

$$\mathbf{Pay}_i = \beta_0 + \beta_1 \mathbf{School}_i + \beta_2 \mathbf{Male}_i + u_i$$

where $\mathbf{School}_i$ gives $i$'s years of schooling; $\mathbf{Male}_i$ represents an indicator variable for whether individual $i$ is male.

Thus

- $\beta_1$: the returns to an additional year of schooling (*ceteris paribus*)
- $\beta_2$: the "premium" for being male (*ceteris paribus*)

If $\beta_2 > 0$, then males are favored in the labor market (discrimination, all else equal).

# Omitted-variable bias

## Example, continued

From our population model

$$\mathbf{Pay}_i = \beta_0 + \beta_1 \mathbf{School}_i + \beta_2 \mathbf{Male}_i + u_i$$

If a study focuses on the relationship between pay and schooling, *i.e.*,

$$\mathbf{Pay}_i = \beta_0 + \beta_1 \mathbf{School}_i + (\beta_2 \mathbf{Male}_i + u_i)$$

$$\mathbf{Pay}_i = \beta_0 + \beta_1 \mathbf{School}_i + \varepsilon_i$$

the "disturbance" becomes $\varepsilon_i = \beta_2 \mathbf{Male}_i + u_i$.

OLS needs exogeneity to be unbiasedness. Likely violated here.

But even if $\boldsymbol{E}[u|X] = 0$, it is not true that $\boldsymbol{E}[\varepsilon|X] = 0$ so long as $\beta_2 \neq 0$.

# Omitted-variable bias

## Example, continued

From our population model

$$\mathbf{Pay}_i = \beta_0 + \beta_1 \mathbf{School}_i + \beta_2 \mathbf{Male}_i + u_i$$

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the "disturbance" becomes $\varepsilon_i = \beta_2 \mathbf{Male}_i + u_i$.

OLS needs exogeneity to be unbiasedness. Likely violated here.

Unless $\mathbf{School}$ and $\mathbf{Male}$ are unrelated, **OLS is biased.**

# Omitted-variable bias

## Example, continued

Let's try to see this result graphically.

Population model:

$$\mathbf{Pay}_i = 20 + 0.5 \times \mathbf{School}_i + 10 \times \mathbf{Male}_i + u_i$$

Our regression model that suffers from omitted-variable bias:

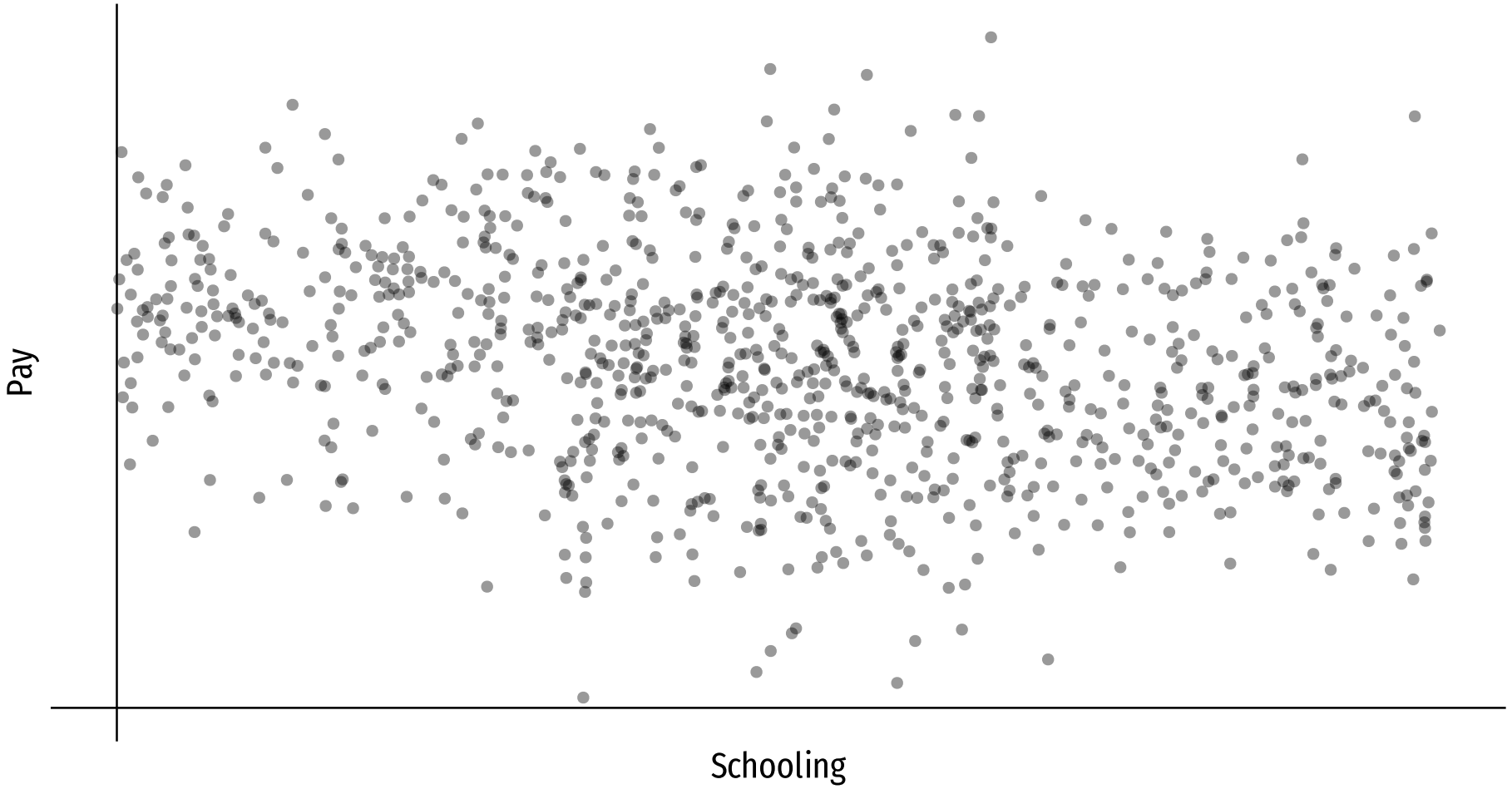
$$\mathbf{Pay}_i = \hat{\beta}_0 + \hat{\beta}_1 \times \mathbf{School}_i + e_i$$

Finally, imagine that women, on average, receive more schooling than men.

# Omitted-variable bias

**Example, continued:** $\text{Pay}_i = 20 + 0.5 \times \text{School}_i + 10 \times \text{Male}_i + u_i$

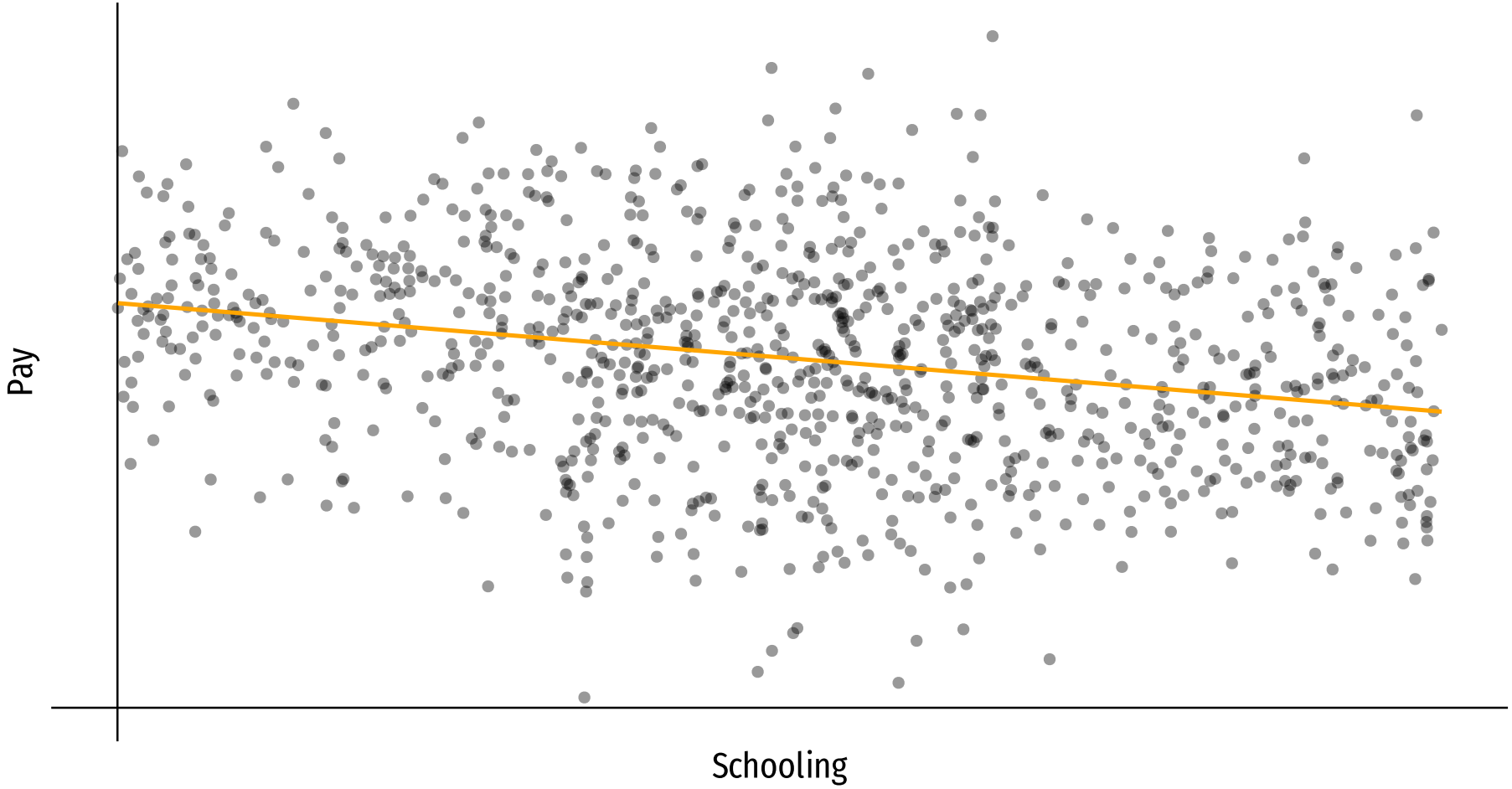
The relationship between pay and schooling.



# Omitted-variable bias

**Example, continued:** $\text{Pay}_i = 20 + 0.5 \times \text{School}_i + 10 \times \text{Male}_i + u_i$

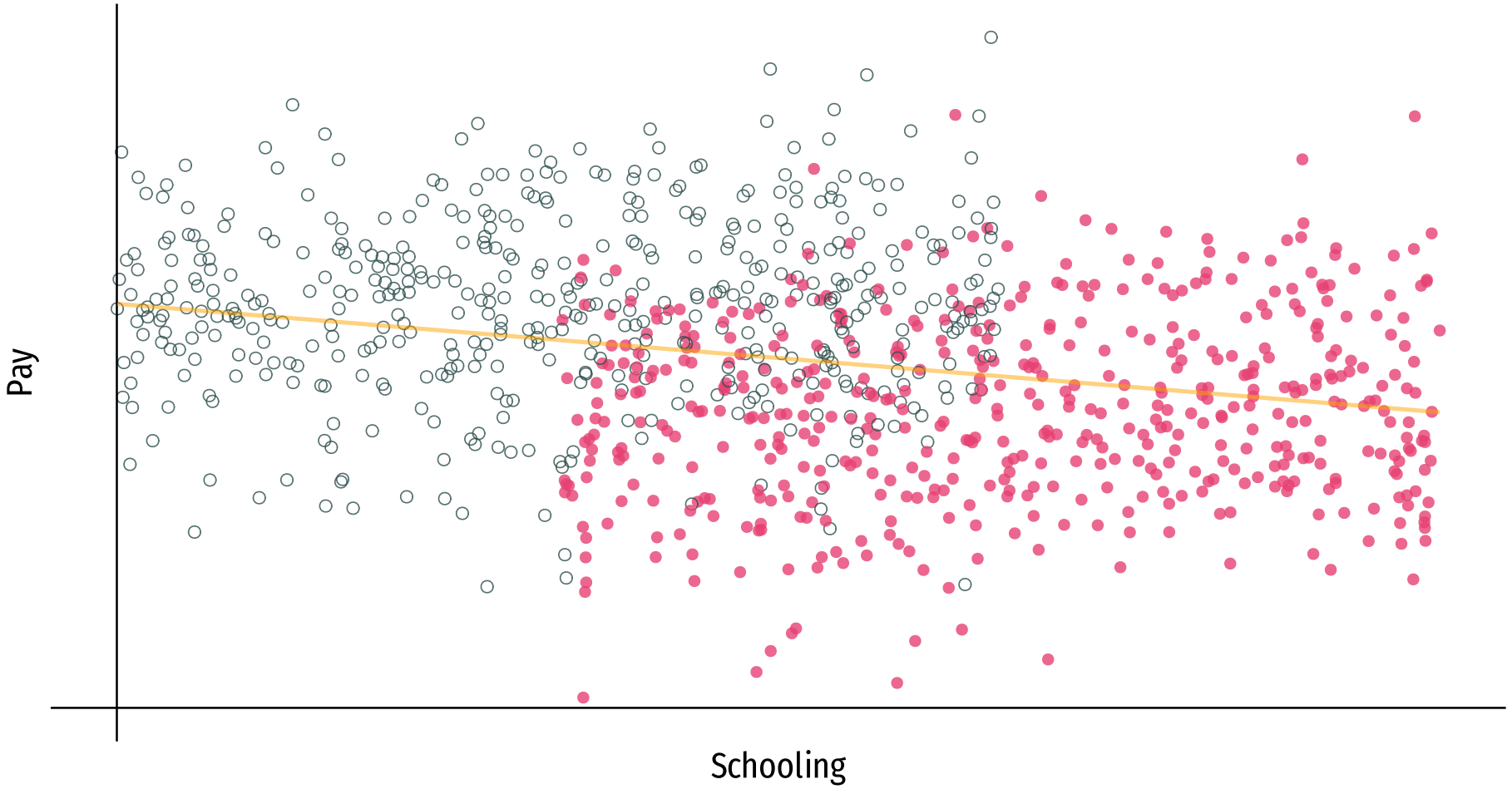
Biased regression estimate: $\widehat{\text{Pay}}_i = 31.3 + -0.9 \times \text{School}_i$



# Omitted-variable bias

**Example, continued:** $\text{Pay}_i = 20 + 0.5 \times \text{School}_i + 10 \times \text{Male}_i + u_i$

Recalling the omitted variable: Gender (**female** and **male**)



# Omitted-variable bias

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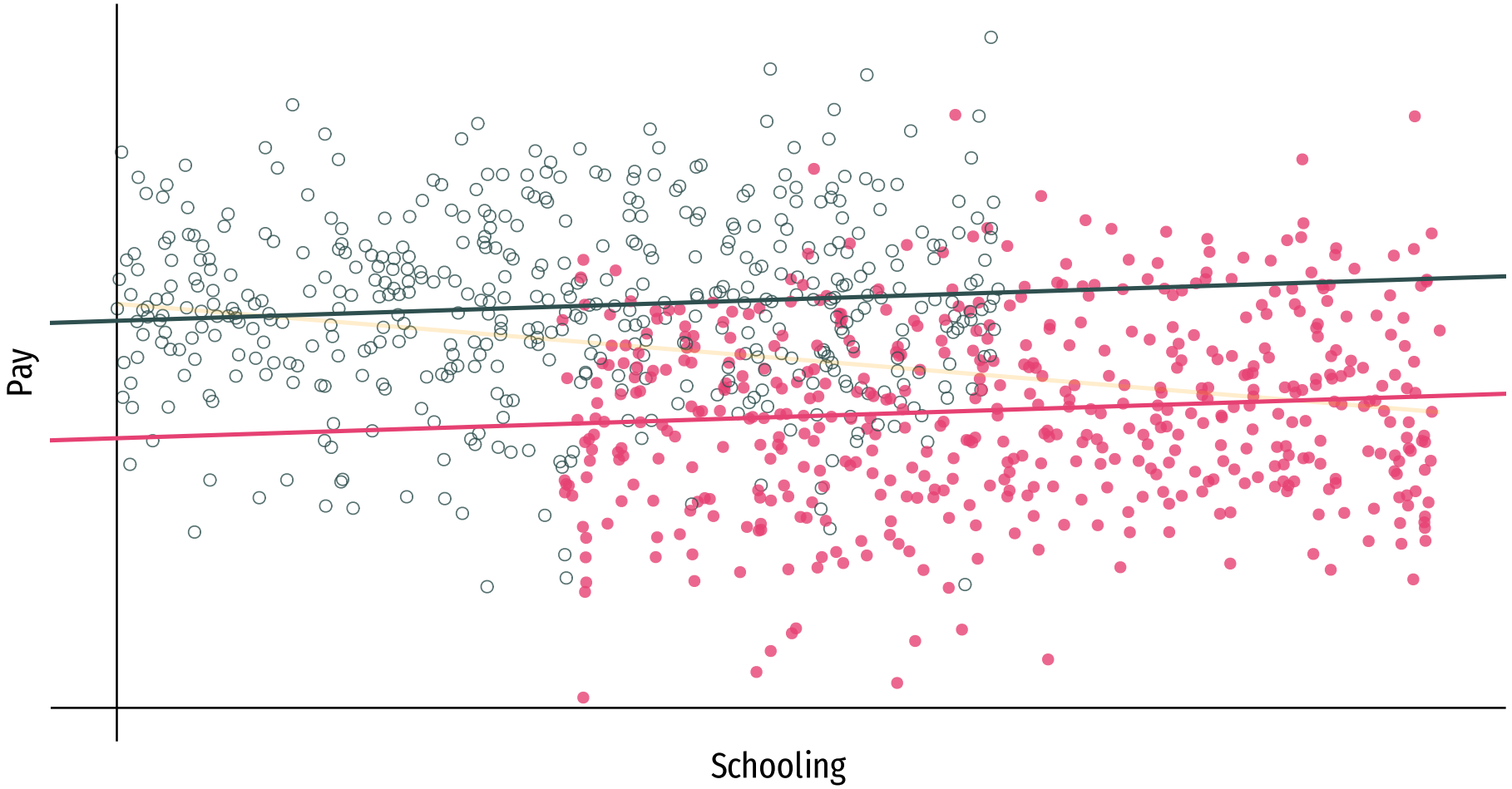
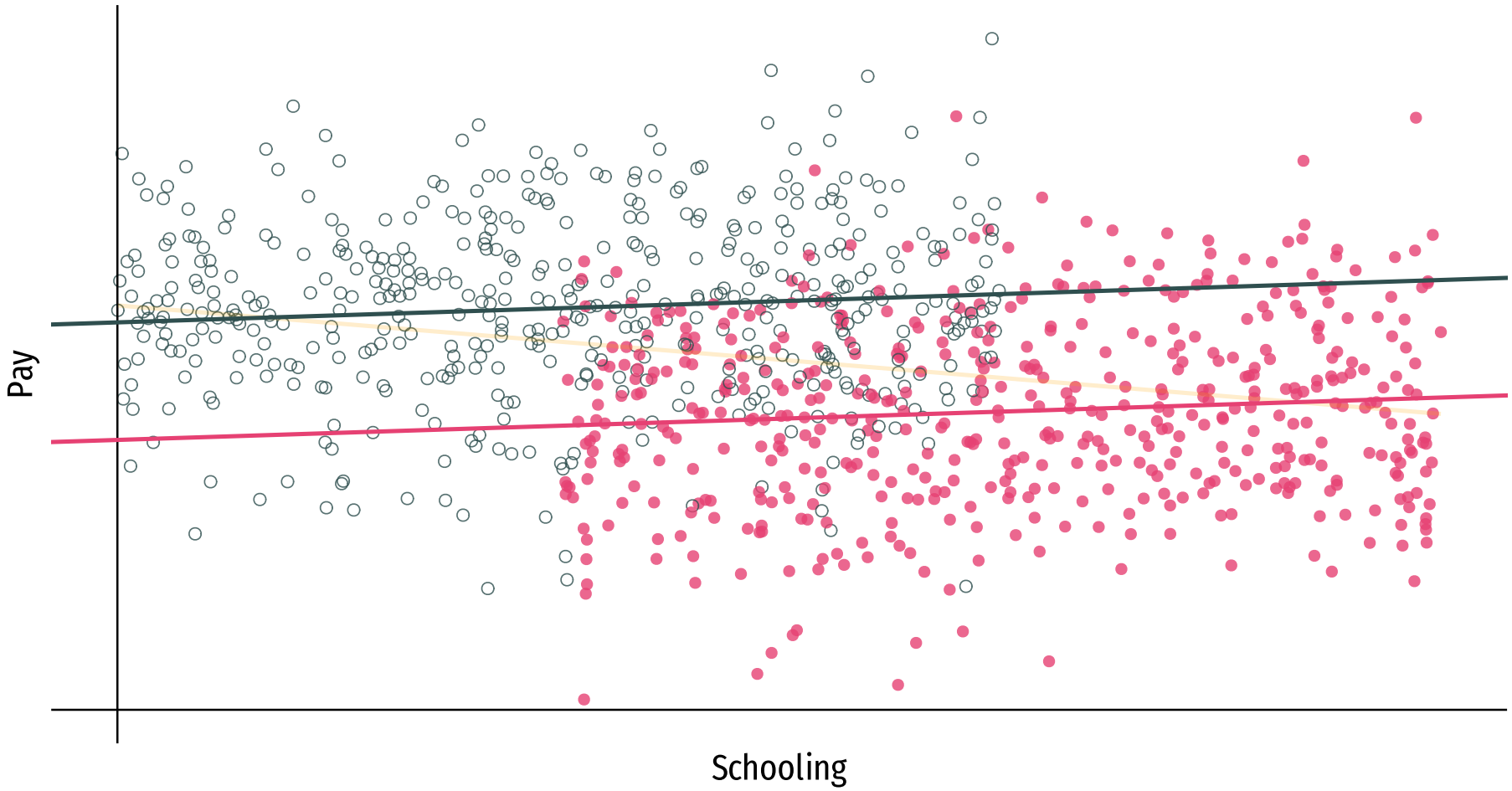


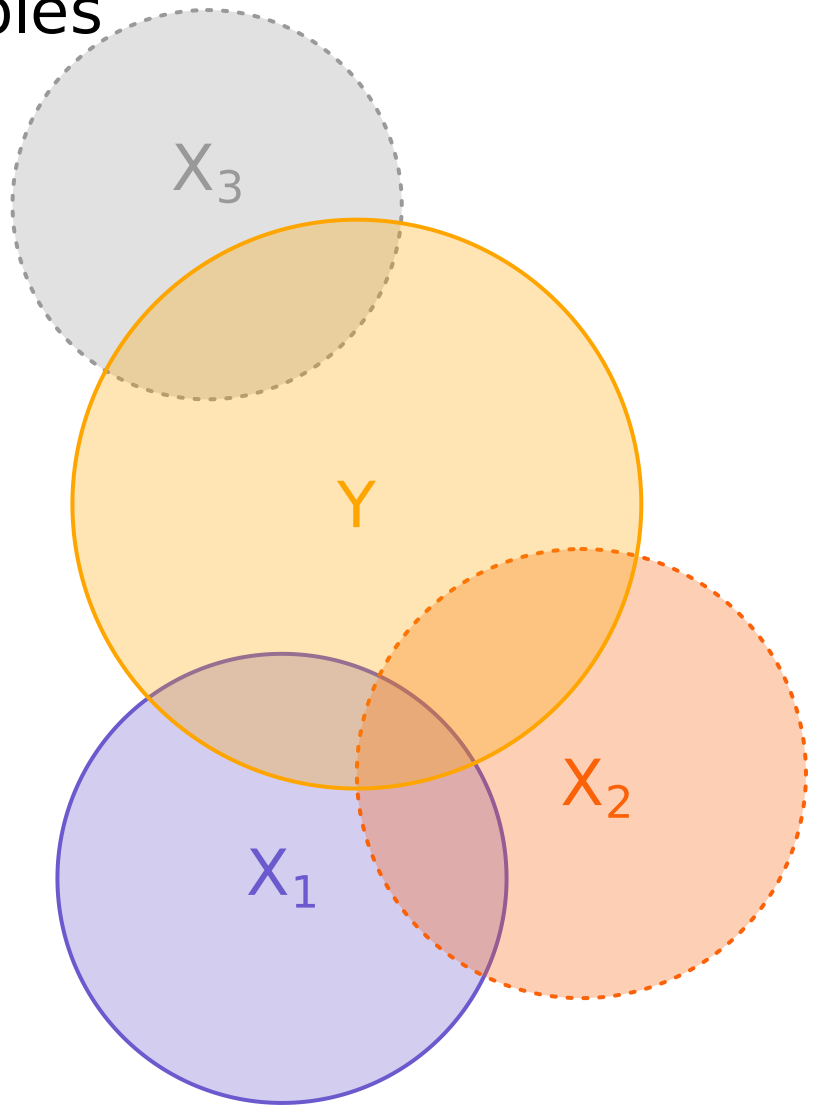
# Omitted-variable bias

**Example, continued:** $\text{Pay}_i = 20 + 0.5 \times \text{School}_i + 10 \times \text{Male}_i + u_i$

Unbiased regression estimate: $\widehat{\text{Pay}}_i = 20.9 + 0.4 \times \text{School}_i + 9.1 \times \text{Male}_i$



# Omitted variables



# Omitted-variable bias

*Solutions*

1. Don't omit variables
2. Instrumental variables and two-stage least squares†

**Warning:** There are situations in which neither solution is possible.

*Interpreting coefficients*

# Interpreting coefficients

*Continuous variables*

Consider the relationship

$$\mathbf{Pay}_i = \beta_0 + \beta_1 \, \mathbf{School}_i + u_i$$

where

- $\mathbf{Pay}_i$ is a continuous variable measuring an individual's pay
- $\mathbf{School}_i$ is a continuous variable that measures years of education

# Interpreting coefficients

*Continuous variables*

Deriving the slope's interpretation:

$$\begin{aligned}
\boldsymbol{E}\left[\text{Pay}|\text{School}=\ell+1\right]-\boldsymbol{E}\left[\text{Pay}|\text{School}=\ell\right]&= \\
\boldsymbol{E}\left[\beta_0+\beta_1(\ell+1)+u\right]-\boldsymbol{E}\left[\beta_0+\beta_1\ell+u\right]&= \\
\left[\beta_0+\beta_1(\ell+1)\right]-\left[\beta_0+\beta_1\ell\right]&= \\
\beta_0-\beta_0+\beta_1\ell-\beta_1\ell+\beta_1&=\beta_1
\end{aligned}$$

# Interpreting coefficients

*Continuous variables*

If we have multiple explanatory variables, *e.g.*,

$$\mathbf{Pay}_i = \beta_0 + \beta_1 \, \mathbf{School}_i + \beta_2 \, \mathbf{Ability}_i + u_i$$

then the interpretation changes slightly.

# Interpreting coefficients

*Continuous variables*

Alternative derivation

Consider the model

$$y = \beta_0 + \beta_1 x + u$$

Differentiate the model:

$$\frac{dy}{dx} = \beta_1$$

# Interpreting coefficients

*Categorical variables*

Consider the relationship

$$\mathbf{Pay}_i = \beta_0 + \beta_1 \, \mathbf{Female}_i + u_i$$

where

- $\mathbf{Pay}_i$ is a continuous variable measuring an individual's pay
- $\mathbf{Female}_i$ is a binary/indicator variable taking $1$ when $i$ is female

# Interpreting coefficients

*Categorical variables*

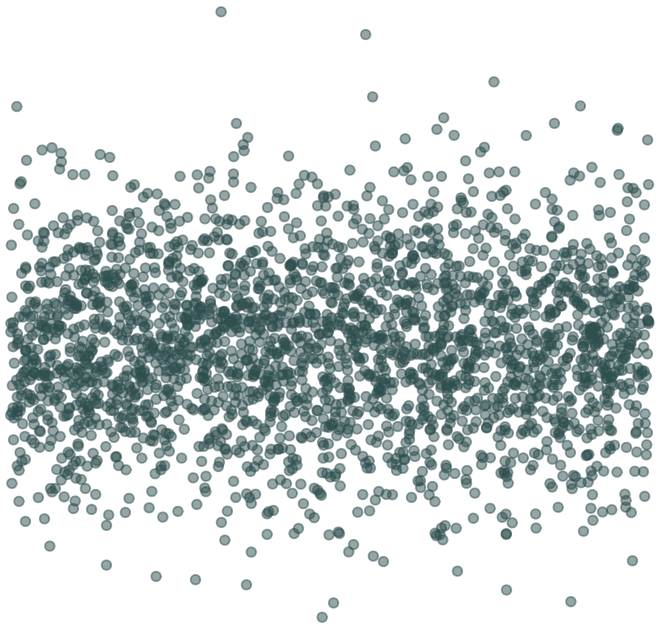
Derivations

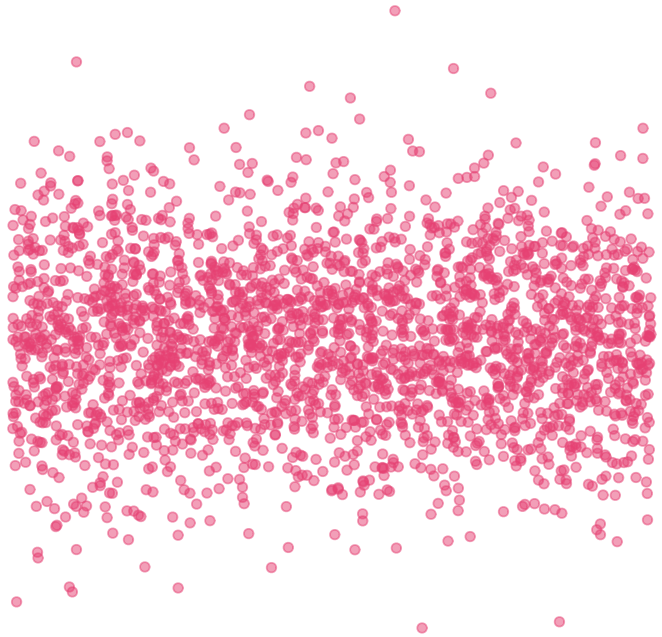
$$
\begin{aligned}
\boldsymbol{E}\left[\text{Pay}|\text{Non-female}\right] &= \boldsymbol{E}\left[\beta_0 + \beta_1 \times 0 + u_i\right] \\
&= \boldsymbol{E}\left[\beta_0 + 0 + u_i\right] \\
&= \beta_0
\end{aligned}
$$

# Interpreting coefficients

*Categorical variables*

$y_i = \beta_0 + \beta_1 x_i + u_i$ for binary variable $x_i = \{0, 1\}$

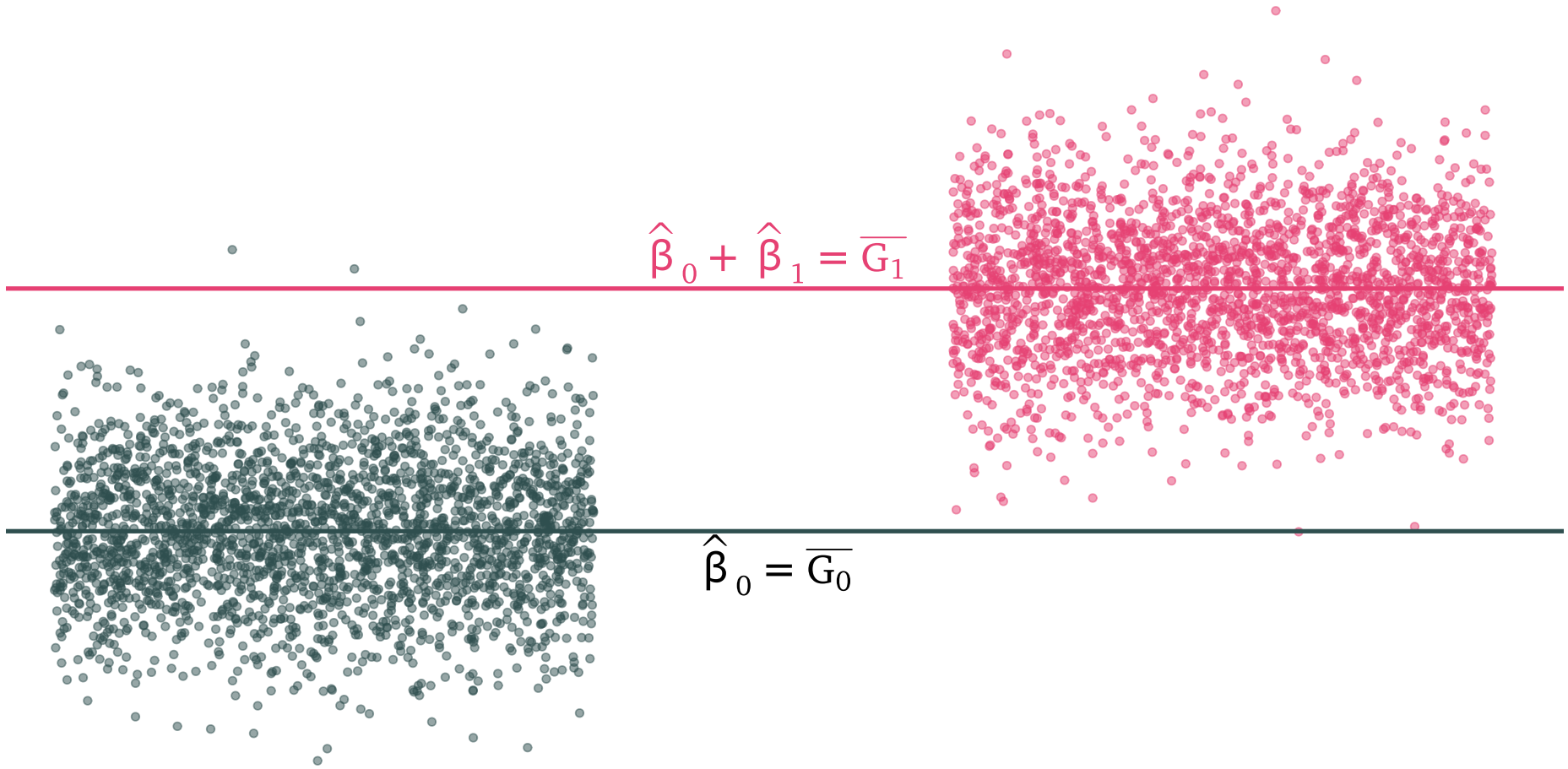




# Interpreting coefficients

*Categorical variables*

$y_i = \beta_0 + \beta_1 x_i + u_i$ for binary variable $x_i = \{0, 1\}$



# Interpreting coefficients

*Interactions*

Interactions allow the effect of one variable to change based upon the level of another variable.

**Examples**

1. Does the effect of schooling on pay change by gender?
2. Does the effect of gender on pay change by race?
3. Does the effect of schooling on pay change by experience?

# Interpreting coefficients

*Interactions*

Previously, we considered a model that allowed women and men to have different wages, but the model assumed the effect of school on pay was the same for everyone:

$$\mathbf{Pay}_i = \beta_0 + \beta_1 \, \mathbf{School}_i + \beta_2 \, \mathbf{Female}_i + u_i$$

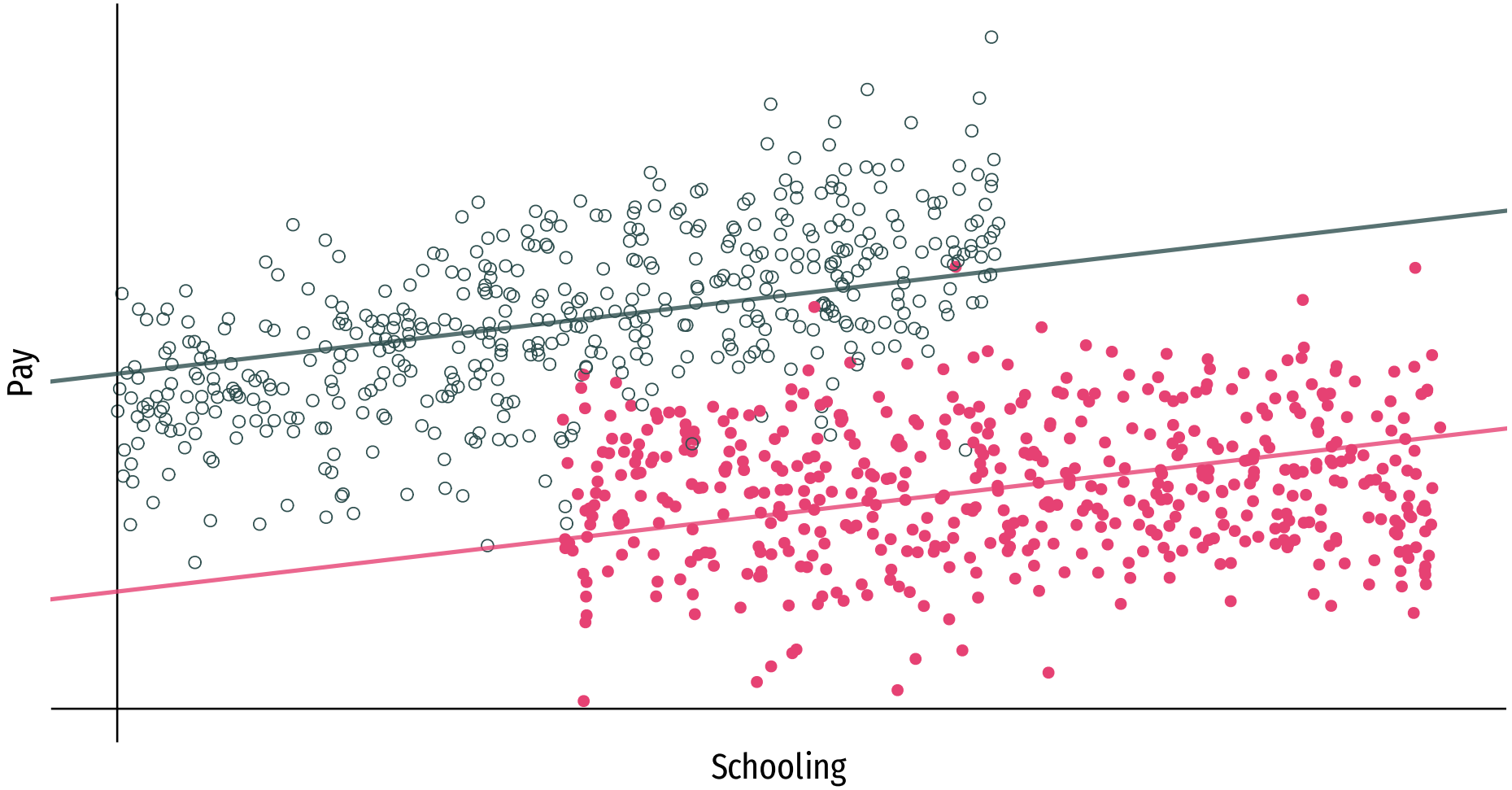
but we can also allow the effect of school to vary by gender:

$$\mathbf{Pay}_i = \beta_0 + \beta_1 \, \mathbf{School}_i + \beta_2 \, \mathbf{Female}_i + \beta_3 \, \mathbf{School}_i \times \mathbf{Female}_i + u_i$$

# Interpreting coefficients

*Interactions*

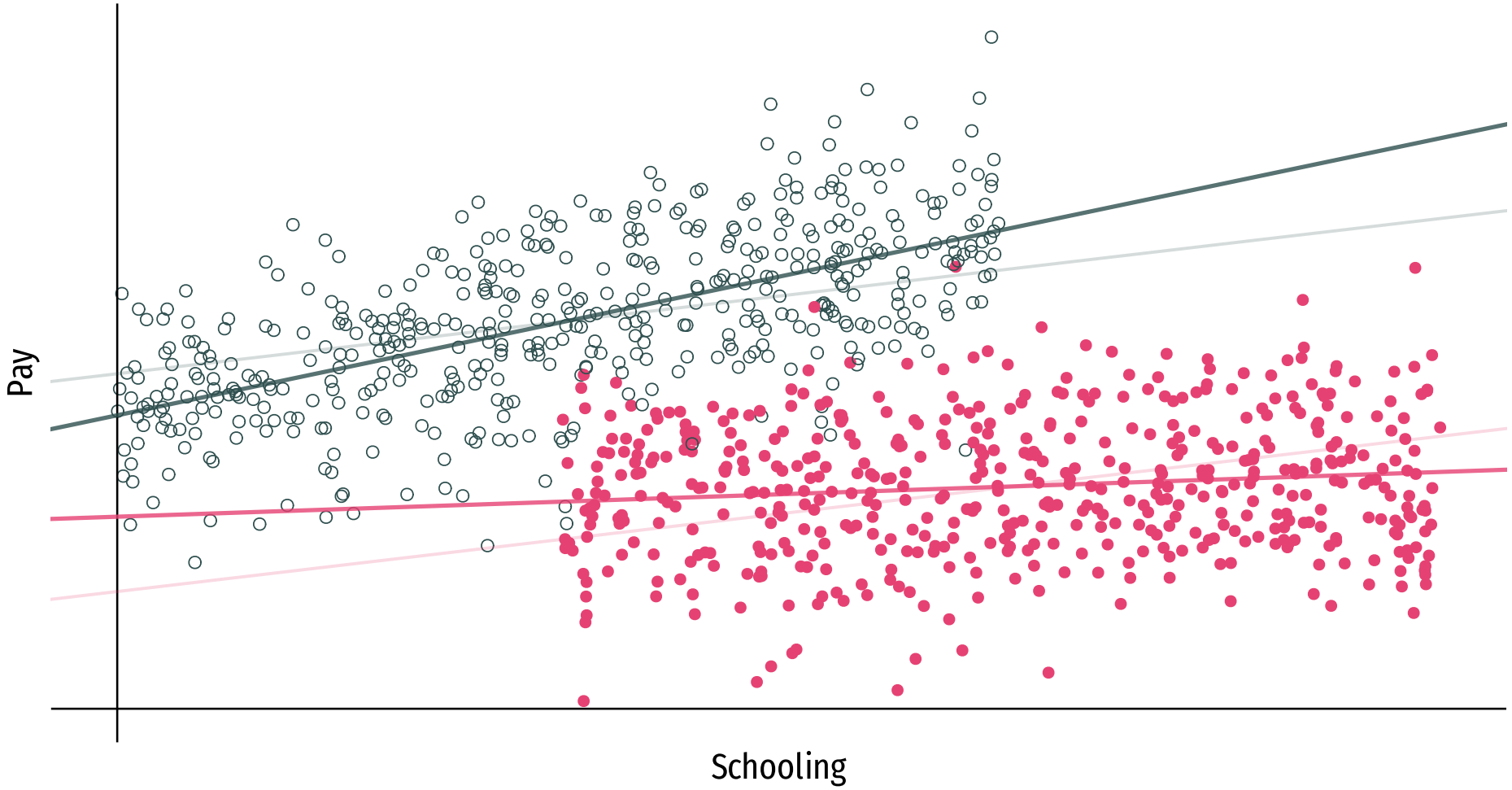
The model where schooling has the same effect for everyone (**F** and **M**):



# Interpreting coefficients

*Interactions*

The model where schooling's effect can differ by gender (**F** and **M**):



# Interpreting coefficients

*Interactions*

Interpreting coefficients can be a little tricky with interactions, but the key is to carefully work through the math[1].

$$\mathbf{Pay}_i = \beta_0 + \beta_1 \, \mathbf{School}_i + \beta_2 \, \mathbf{Female}_i + \beta_3 \, \mathbf{School}_i \times \mathbf{Female}_i + u_i$$

Expected returns for an additional year of schooling for women:

$$\begin{aligned} \boldsymbol{E}\left[\mathbf{Pay}_i | \mathbf{Female} \wedge \mathbf{School} = \ell + 1\right] - \boldsymbol{E}\left[\mathbf{Pay}_i | \mathbf{Female} \wedge \mathbf{School} = \ell\right] = \\ \boldsymbol{E}\left[\beta_0 + \beta_1(\ell + 1) + \beta_2 + \beta_3(\ell + 1) + u_i\right] - \boldsymbol{E}\left[\beta_0 + \beta_1 \ell + \beta_2 + \beta_3 \ell + u_i\right] = \\ \beta_1 + \beta_3 \end{aligned}$$

1. As is often the case with econometrics.

# Interpreting coefficients

*Interactions*

Interpreting coefficients can be a little tricky with interactions, but the key is to carefully work through the math[1].

$$\mathbf{Pay}_i = \beta_0 + \beta_1 \, \mathbf{School}_i + \beta_2 \, \mathbf{Female}_i + \beta_3 \, \mathbf{School}_i \times \mathbf{Female}_i + u_i$$

Expected returns for an additional year of schooling for women: $\beta_1 + \beta_3$

- $\beta_1$: the expected return for an add. yr. of schooling for non-females;
- $\beta_3$: the **difference in the returns to schooling** for females *vs.* non-females.

1. As is often the case with econometrics.

# Interpreting coefficients

*Interactions*

The previous slides focused on interactions where one variable was **binary**.

If both variables are continuous, then the interpretation is slightly trickier.

*Key* Interactions simply mean the effect of one variable depends on the level of another variable.

# Interpreting coefficients

*Interactions*

Suppose we're interested in the model

$$\mathbf{Pay}_i = \beta_0 + \beta_1 \, \mathbf{School}_i + \beta_2 \, \mathbf{Experience}_i + \beta_3 \, \mathbf{School}_i \times \mathbf{Experience}_i + u_i$$

where $\mathbf{School}_i$ and $\mathbf{Experience}_i$ are both continuous variables (in years).

# Interpreting coefficients

*Interactions*

In the model

$$\mathbf{Pay}_i = \beta_0 + \beta_1 \, \mathbf{School}_i + \beta_2 \, \mathbf{Experience}_i + \beta_3 \, \mathbf{School}_i \times \mathbf{Experience}_i + u_i$$

all else equal, an additional year of school changes pay by

$$\beta_1 + \beta_3 \mathbf{Experience}$$

# Interpreting coefficients

*Polynomials*

Polynomials are just interactions: they interact a variable with itself.

$$\text{Pay}_i = \beta_0 + \beta_1 \text{School}_i + \beta_2 \text{School}_i^2 + u_i$$

Here the effect of schooling depends on an individual's level of schooling.

**Interpretation** Back to the partial derivative:

$$\frac{\partial \text{Pay}_i}{\partial \text{School}_i} = \beta_1 + 2\beta_2 \text{School}_i$$

all else equal, an additional year of school changes pay by

$$\beta_1 + 2\beta_2 \text{School}_i$$

# Interpreting coefficients

*Binary outcomes*

When your **outcome variable** is binary, the interpretation changes slightly.

*Recall:* The avg. of a binary variable gives the % of observations with a '1'.

*Example:* Avg(0, 0, 0, 1, 1) = 0.40 $\implies$ 40% of obserations = 1.

# Interpreting coefficients

*Log-linear specification*

In economics, you will frequently see logged outcome variables with linear (non-logged) explanatory variables, *e.g.*,

$$\log(\text{Pay}_i) = \beta_0 + \beta_1 \, \text{School}_i + u_i$$

This specification changes our interpretation of the slope coefficients.

**Interpretation**

- A one-unit increase in our explanatory variable increases the outcome variable by approximately $\beta_1 \times 100$ percent.
- *Example:* An additional year of schooling increases pay by approximately 3 percent (for $\beta_1 = 0.03$).

# Interpreting coefficients

*Log-linear specification*

## Derivation

Consider the log-linear model

$$\log(y) = \beta_0 + \beta_1 \, x + u$$

and differentiate

$$\frac{dy}{y} = \beta_1 dx$$

So a marginal change in $x$ (*i.e.*, $dx$) leads to a $\beta_1 dx$ **percentage change** in $y$.

# Interpreting coefficients

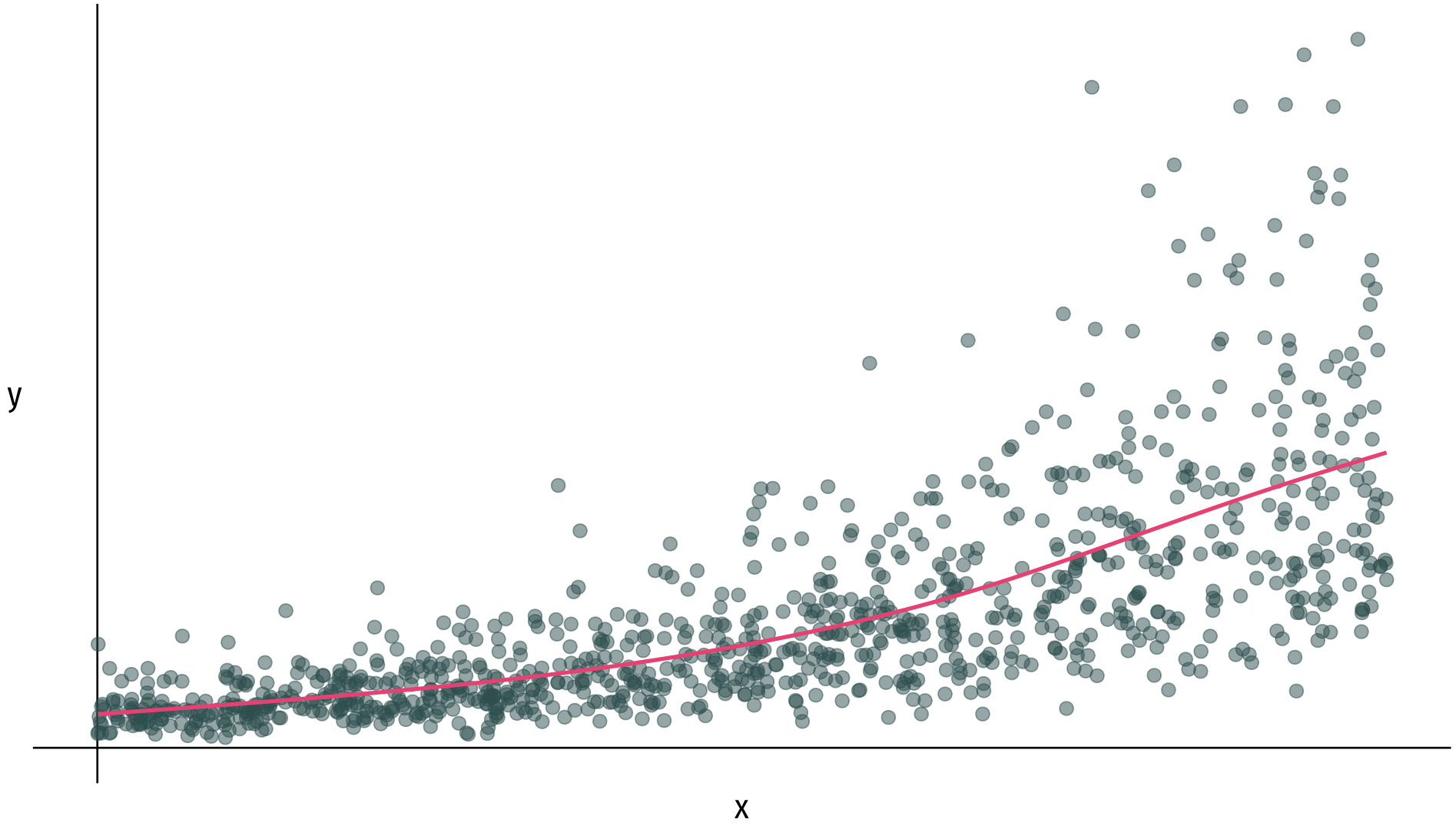
*Log-linear specification*

Because the log-linear specification comes with a different interpretation, you need to make sure it fits your data-generating process/model.

Does $x$ change $y$ in levels (*e.g.*, a 3-unit increase) or percentages (*e.g.*, a 10-percent increase)?

# Interpreting coefficients

*Log-linear specification*



# Interpreting coefficients

*Log-log specification*

Similarly, econometricians frequently employ log-log models, in which the outcome variable is logged *and* at least one explanatory variable is logged

$$\log(\text{Pay}_i) = \beta_0 + \beta_1 \log(\text{School}_i) + u_i$$

**Interpretation:**

- A one-percent increase in $x$ will lead to a $\beta_1$ percent change in $y$.
- Often interpreted as an elasticity.

# Interpreting coefficients

*Log-log specification*

## Derivation

Consider the log-log model

$$\log(y) = \beta_0 + \beta_1 \log(x) + u$$

and differentiate

$$\frac{dy}{y} = \beta_1 \frac{dx}{x}$$

which says that for a one-percent increase in $x$, we will see a $\beta_1$ percent increase in $y$. As an elasticity:

$$\frac{dy}{dx}\frac{x}{y} = \beta_1$$

# Interpreting coefficients

*Log-linear with a binary variable*

**Note:** If you have a log-linear model with a binary indicator variable, the interpretation for the coefficient on that variable changes.

Consider

$$\log(y_i) = \beta_0 + \beta_1 x_1 + u_i$$

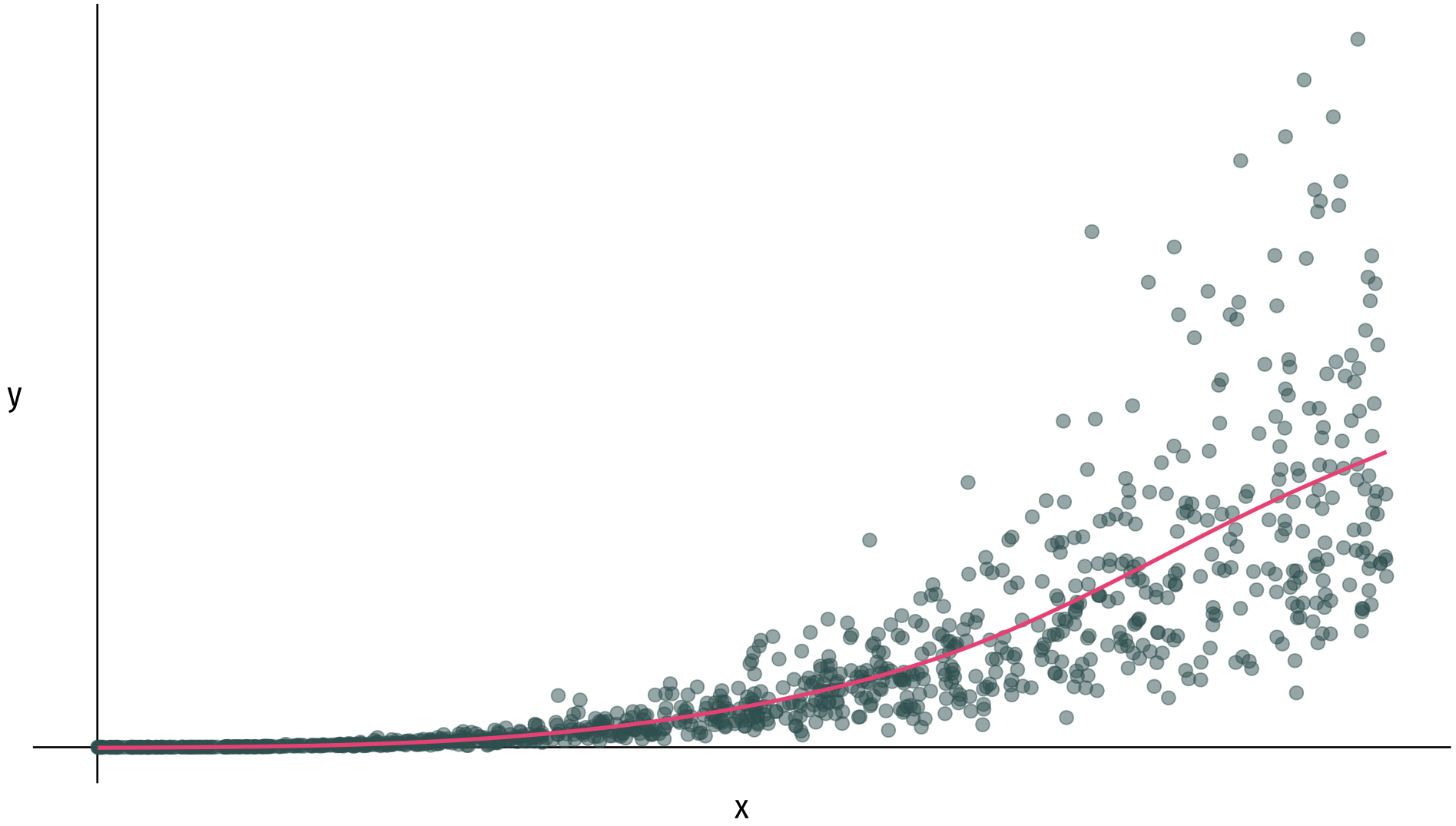
for binary variable $x_1$.

The interpretation of $\beta_1$ is now

- When $x_1$ changes from 0 to 1, $y$ will change by $100 \times \left(e^{\beta_1} - 1\right)$ percent.
- When $x_1$ changes from 1 to 0, $y$ will change by $100 \times \left(e^{-\beta_1} - 1\right)$ percent.

# Interpreting coefficients

*Log-log specification*



*Additional topics*

# Additional topics

*Inference vs. prediction*

So far, we've focused mainly **statistical (causal) inference**—using estimators and their distributions properties to try to learn about underlying, unknown population parameters.

$$y_i = \hat{\beta}_0 + \hat{\beta}_1 x_{1i} + \hat{\beta}_2 x_{2i} + \cdots + \hat{\beta}_k x_{ki} + e_i$$

# Additional topics

*Inference vs. prediction*

Succinctly

- **Inference:** causality, $\hat{\beta}_k$ (consistent and efficient), standard errors/hypothesis tests for $\hat{\beta}_k$, generally OLS
- **Prediction:**[1] correlation, $\hat{y}_i$ (low error), model selection, nonlinear models are much more common

1. Includes forecasting.

# Additional topics

*Treatment effects and causality*

Much of modern (micro)econometrics focuses on causally estimating (*identifying*) the effect of programs/policies, *e.g.*,

- Government shutdowns
- The minimum wage
- Recreational-cannabis legalization
- Salary-history bans
- Preschool
- The Clean Water Act

# Additional topics

*Transformations*

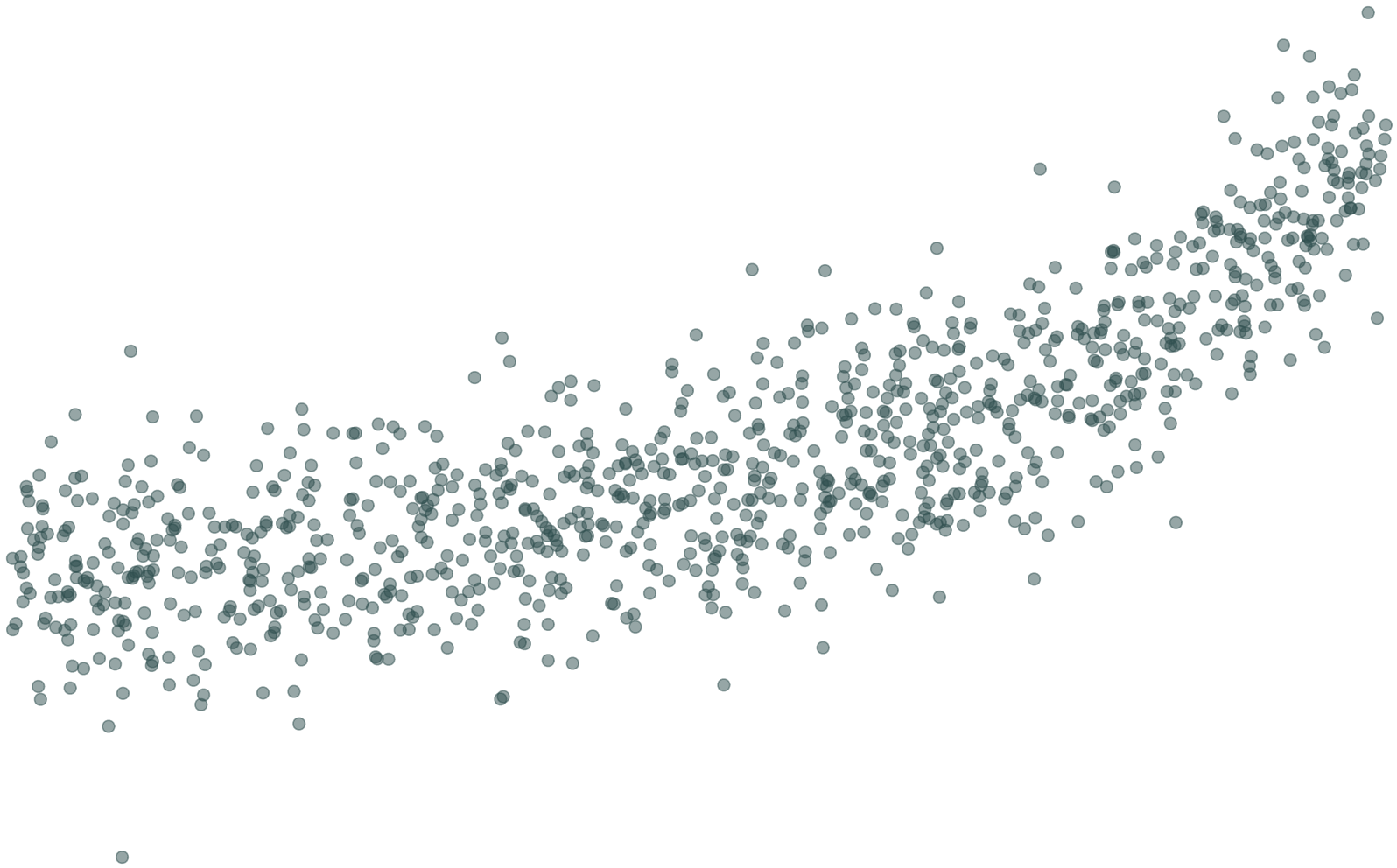
Our linearity assumption requires

1. **parameters enter linearly** (*i.e.*, the $\beta_k$ multiplied by variables)
2. the $u_i$ **disturbances enter additively**

We allow nonlinear relationships between $y$ and the explanatory variables.

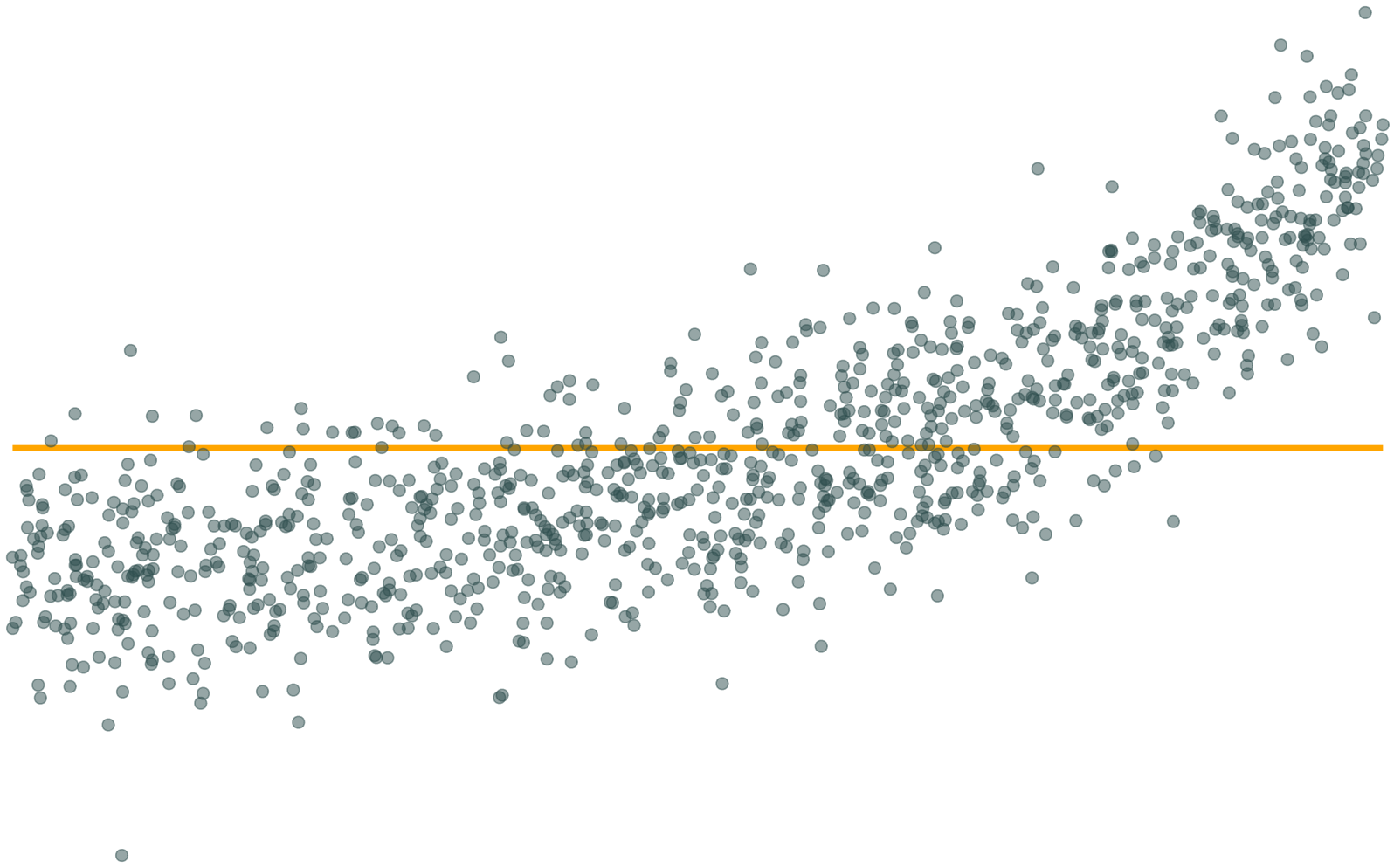
# Additional topics

**Transformation challenge:** (literally) infinite possibilities. What do we pick?



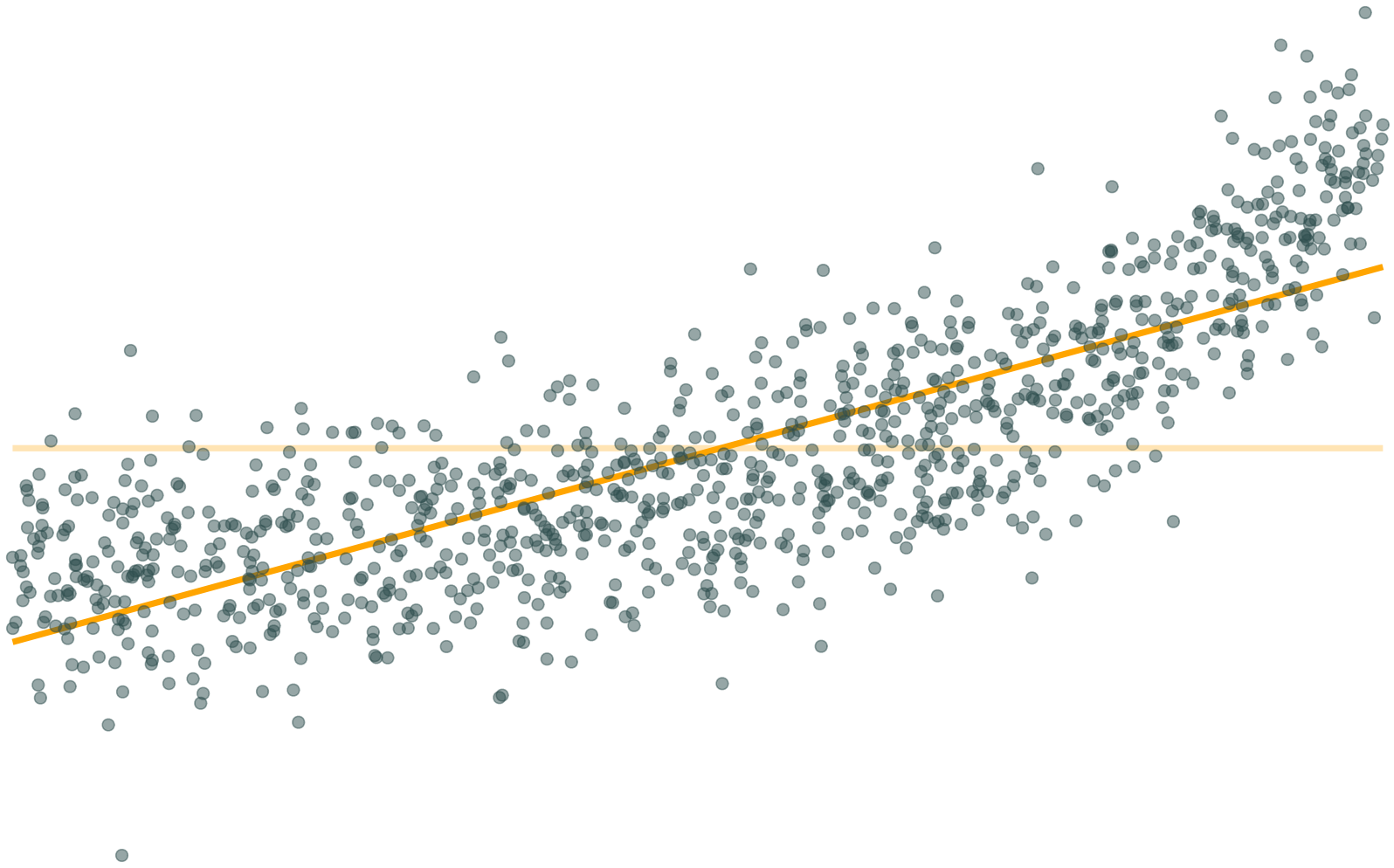
# Additional topics

$y_i = \beta_0 + u_i$



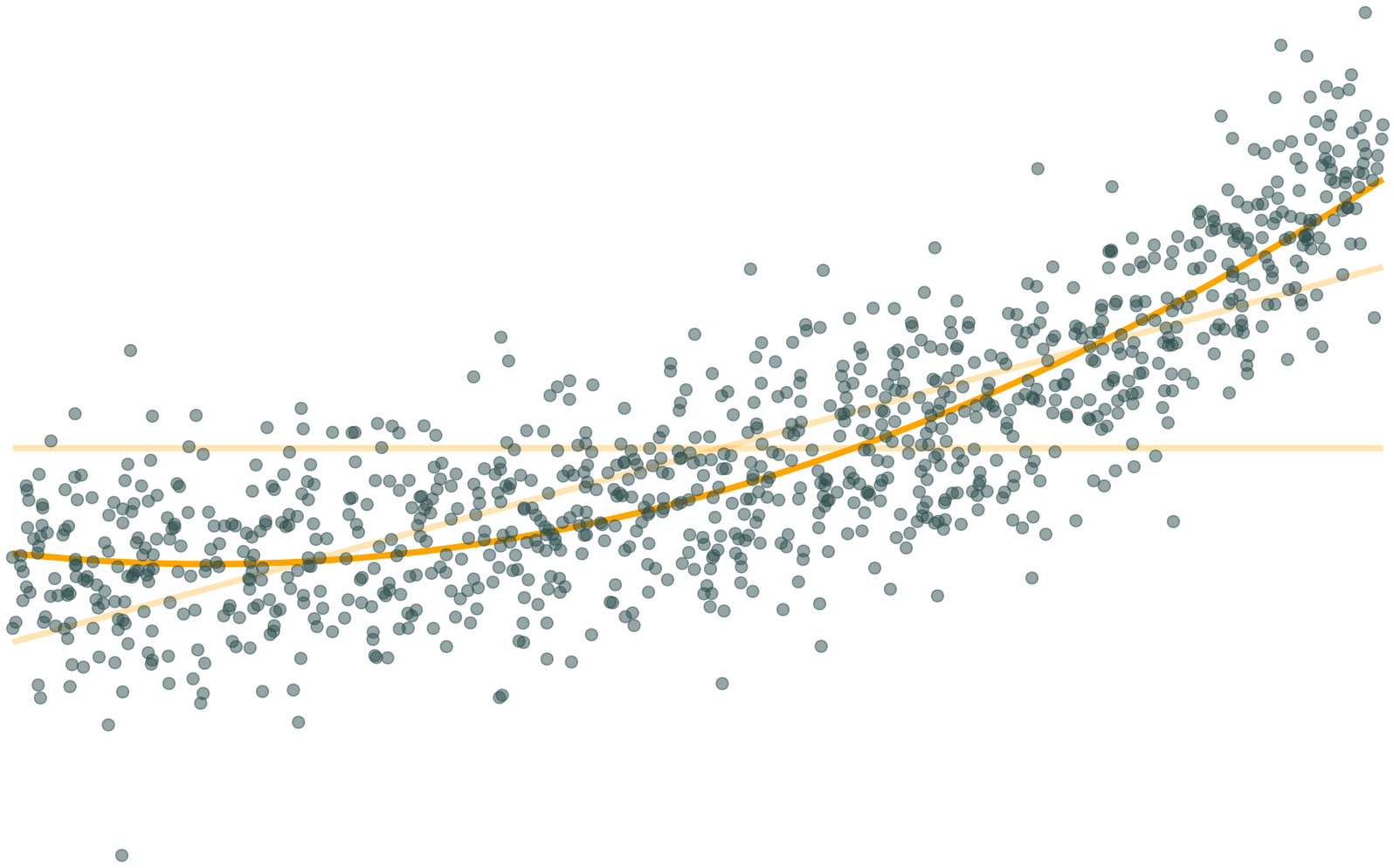
# Additional topics

$$y_i = \beta_0 + \beta_1 x + u_i$$



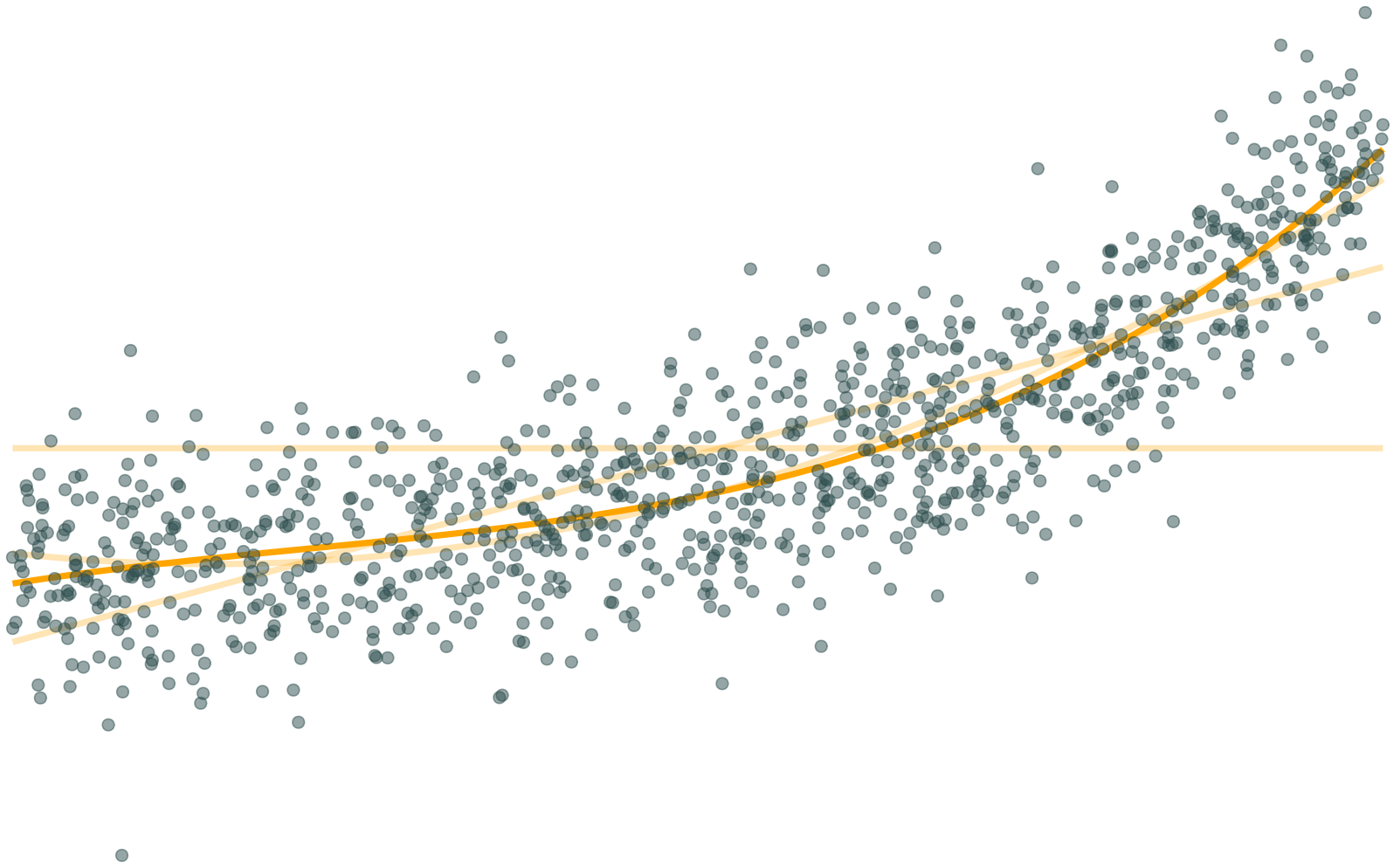
# Additional topics

$$y_i = \beta_0 + \beta_1 x + \beta_2 x^2 + u_i$$



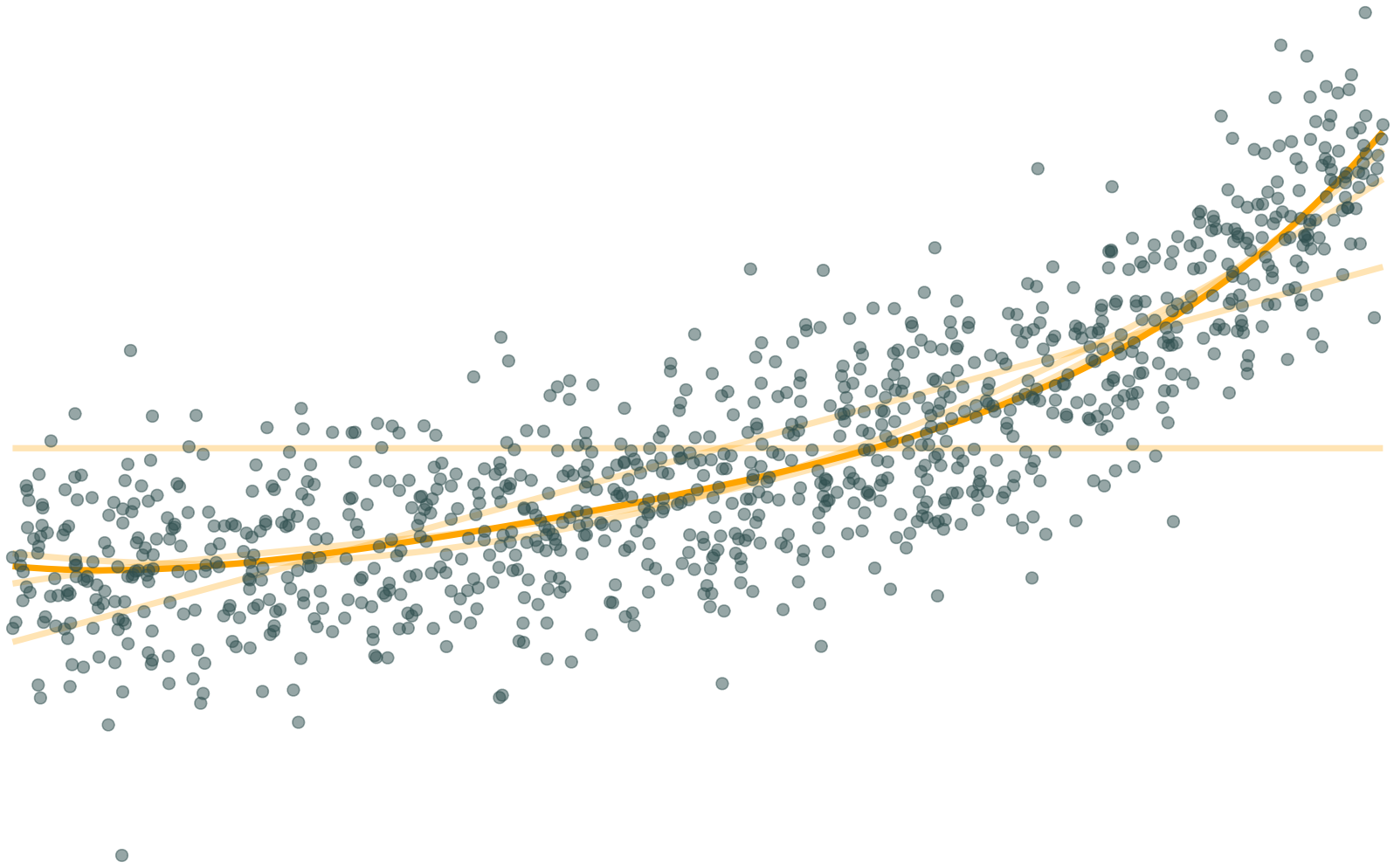
# Additional topics

$$y_i = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + u_i$$



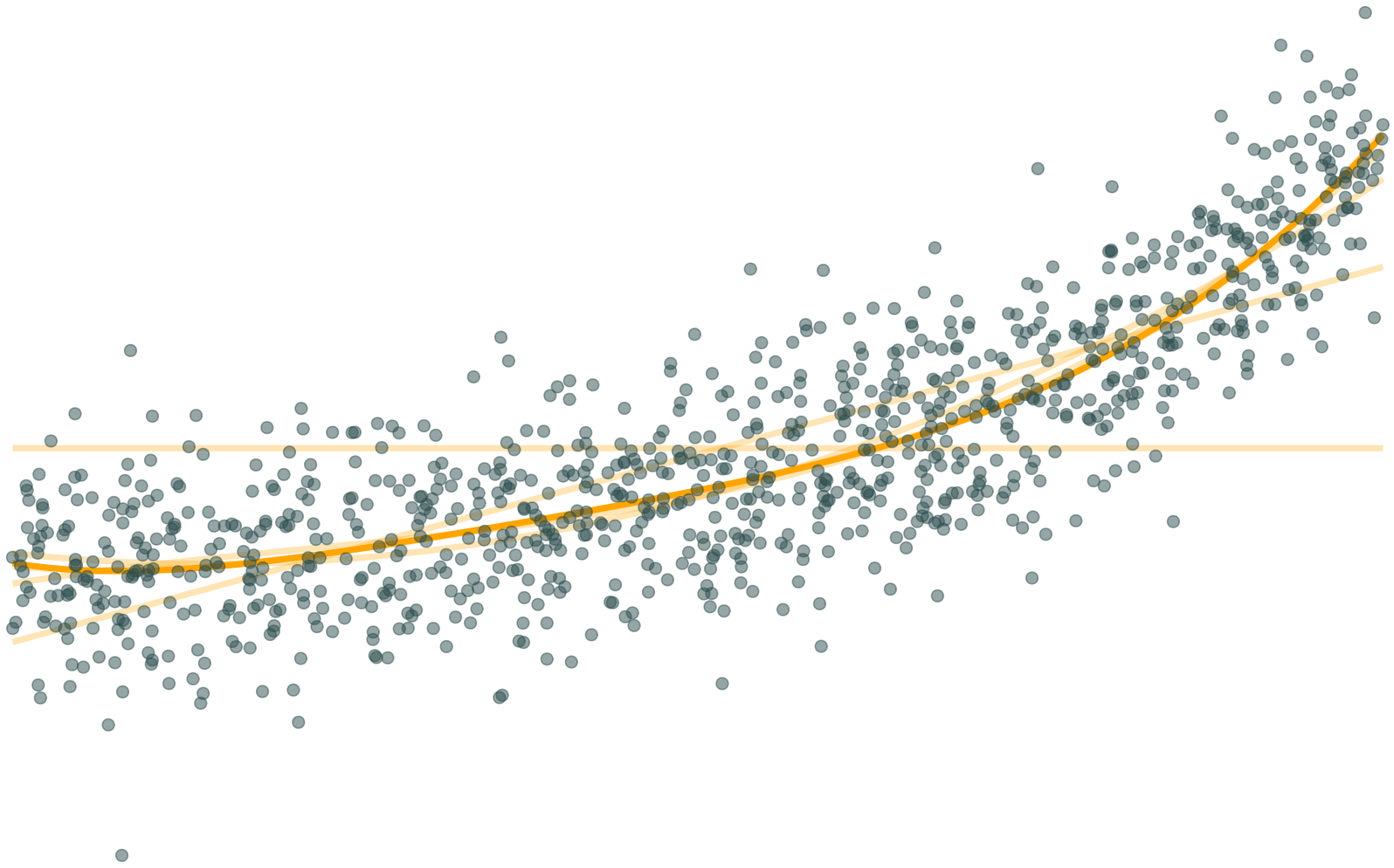
# Additional topics

$$y_i = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 x^4 + u_i$$



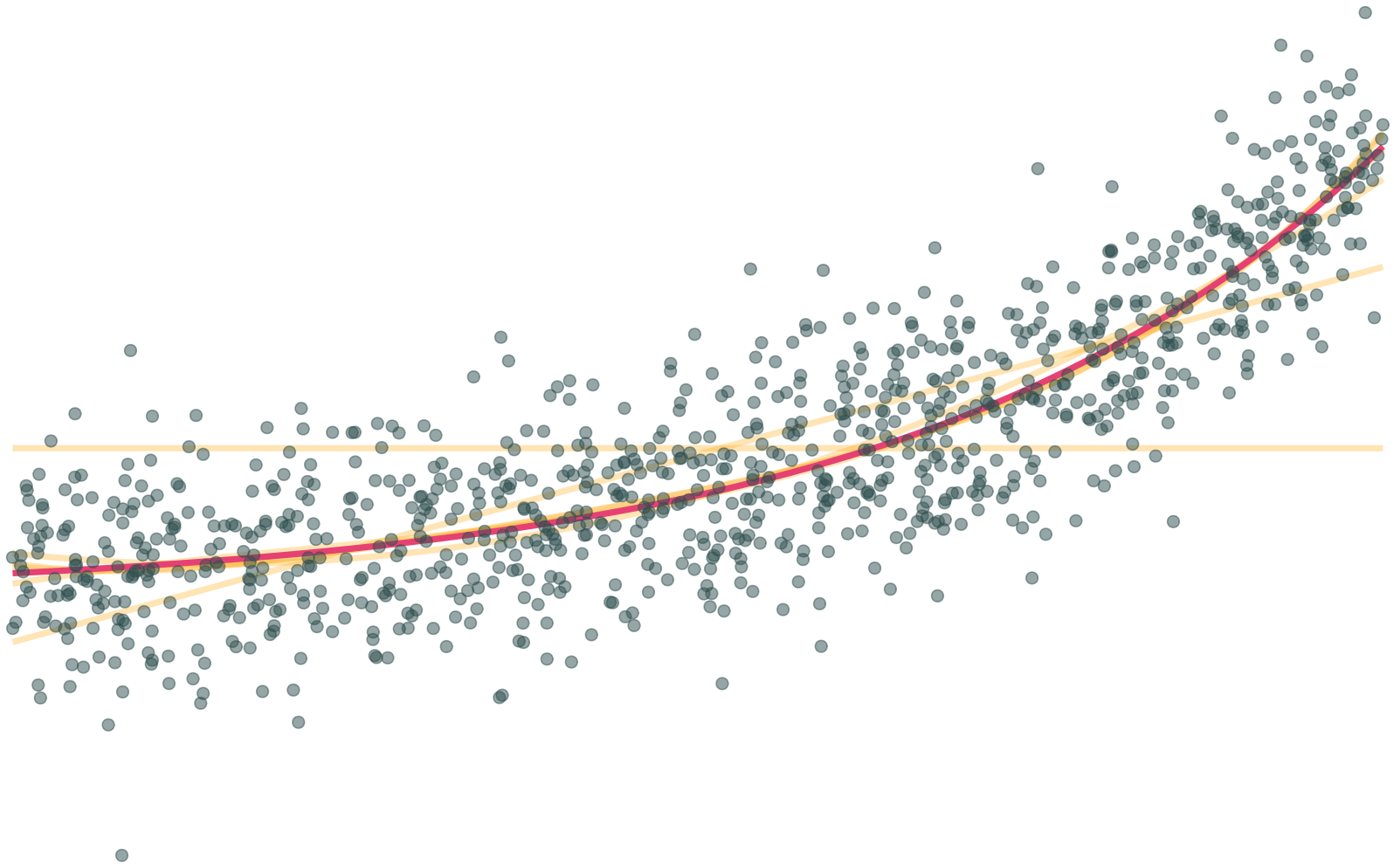
# Additional topics

$$y_i = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 x^4 + \beta_5 x^5 + u_i$$



# Additional topics

**Truth:** $y_i = 2e^x + u_i$



# Additional topics

*Outliers*

Because OLS minimizes the sum of the **squared** errors, outliers can play a large role in our estimates.

**Common responses**

- Remove the outliers from the dataset
- Replace outliers with the $99^{th}$ percentile of their variable (*Windsorize*)
- Take the log of the variable to "take care of" outliers
- Do nothing. Outliers are not always bad. Some people are "far" from the average. It may not make sense to try to change this variation.

# Additional topics

*Missing data*

Similarly, missing data can affect your results.

R doesn't know how to deal with a missing observation.

```
#> [1] NA
```

If you run a regression[1] with missing values, R drops the observations missing those values.

If the observations are missing in a nonrandom way, a random sample may end up nonrandom.

1. Or perform almost any operation/function.

# Wrapping up

# Wrapping up

We've refresshed the main ingredients for OLS regression:

- OLS estimates unknown **population** parameters using a random **sample**; uncertainty is unavoidable.
- standard errors, conf. intervals, and hyp. tests help us **infer** from sample to population (while accounting for sampling variability/uncertainty).
- **omitted variables** bias $\beta$ est. when they affect $y$ & corr. w/ regressors.
- coef. **interpretation** depends on the variable type & model spec.

# Wrapping up

*Moving forward...*

So far, the big message has been

- OLS is powerful when we satisfy its assumptions.
- Different assumptions protect different results
  - ↳ exogeneity provides for unbiased-ness,
  - ↳ disturbance var./cov. affects for efficiency and inference.
- Residuals $e_i$ help us learn about unobserved disturbances $u_i$.

*Next* What happens when we violate $u_i$ var. or cov. assumptions?

- Is OLS biased? What happens to efficiency and inference?
- What can we do about it?